

2-8-8

A particle starts at time $t=0$ and moves along the x -axis so that its position at any time $t \geq 0$ is given by $x(t)=(t-1)^3(2t-3)$.

1. Find the value of t when the particle is moving and the acceleration is zero.

Part C

1 point is earned for derivative of their v from (a) or (b)

1 point is earned for solving for $v'(t)=0$ and solves for at least 2 roots

1 point is earned for answer

$$\begin{aligned} a(t) &= v'(t) \\ &= 2(t-1)(8t-11) + 8(t-1)^2 \\ &= 6(t-1)(4t-5) \end{aligned}$$

$$a(t) = 0 \text{ when } t = 1, t = \frac{5}{4}$$

but particle not moving at $t = 1$ so $t = \frac{5}{4}$



0	1	2	3
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The student response earns three of the following points:

1 point is earned for derivative of their v from (a) or (b)

1 point is earned for solving for $v'(t)=0$ and solves for at least 2 roots

1 point is earned for answer

$$\begin{aligned} a(t) &= v'(t) \\ &= 2(t-1)(8t-11) + 8(t-1)^2 \\ &= 6(t-1)(4t-5) \end{aligned}$$

$$a(t) = 0 \text{ when } t = 1, t = \frac{5}{4}$$

but particle not moving at $t = 1$ so $t = \frac{5}{4}$

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2. Find the velocity of the particle at any time $t \geq 0$.
-

Part A

2 points are earned for the derivative

1 point is **deducted** for algebra error

$$\begin{aligned}v(t) &= x'(t) \\&= 3(t-1)^2(2t-3) + 2(t-1)^3 \\&= (t-1)^2(8t-11)\end{aligned}$$

✓

0	1	2
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The student response earns two of the following points:

2 points are earned for the derivative

1 point is **deducted** for algebra error

$$\begin{aligned}v(t) &= x'(t) \\&= 3(t-1)^2(2t-3) + 2(t-1)^3 \\&= (t-1)^2(8t-11)\end{aligned}$$

2-8-8

3. Suppose that the function f has a continuous second derivative for all x , and that $f(0) = 2$, $f'(0) = -3$, and $f''(0) = 0$. Let g be a function whose derivative is given by $g'(x) = e^{-2x} (3f(x) + 2f'(x))$ for all x .

Show that $g''(x) = e^{-2x} (-6f(x) - f'(x) + 2f''(x))$. Does g have a local maximum at $x = 0$? Justify your answer.

Part D

2 points are earned for verify derivative 0/2 product or chain rule error

< -1 > algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x}) (3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) - f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for verify derivative 0/2 product or chain rule error

< -1 > algebra errors

1 point is earned for: $g'(0) = 0$ and $g''(0)$

1 point is earned for the answer and reasoning

$$g'(x) = e^{-2x} (3f(x) + 2f'(x))$$

$$g''(x) = (-2e^{-2x}) (3f(x) + 2f'(x)) + e^{-2x} (3f'(x) + 2f''(x))$$

$$= e^{-2x} (-6f(x) - f'(x) + 2f''(x))$$

$$g''(0) = e^0 [(-6)(2) - (-3) + 2(0)] = -9$$

Since $g'(0) = 0$ and $g''(0) < 0$, g does have a local maximum at $x = 0$.

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4. Let F be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .

Find the x -coordinate of each critical point of F . For each such x , determine whether $f(x)$ is a relative maximum, a relative minimum, or neither.

Part B

1 point is earned for correct derivative

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x=1$

relative maximum at $x=1$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for correct derivative

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

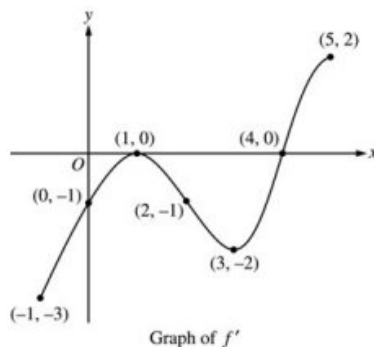
$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x=1$

relative maximum at $x=1$

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5.



The figure above shows the graph of f' , the derivative of the function f , on the closed interval $-1 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = 1$ and $x = 3$. The function f is twice differentiable with $f(2) = 6$.

- (a) Find the x -coordinate of each of the points of inflection of the graph of f . Give a reason for your answer.
- (b) At what value of x does f attain its absolute minimum value on the closed interval $-1 \leq x \leq 5$? At what value of x does f attain its absolute maximum value on the closed interval $-1 \leq x \leq 5$? Show the analysis that leads to your answers.
- (c) Let g be the function defined by $g(x) = xf(x)$. Find an equation for the line tangent to the graph of g at $x = 2$.

Part A

1 point is earned for $x = 1, x = 3$

1 point is earned for the reason

$x = 1$ and $x = 3$ because the graph of f' changes from increasing to decreasing at $x = 1$, and changes from decreasing to increasing at $x = 3$.

✓

0	1	2
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The student response earns all of the following points:

1 point is earned for $x = 1, x = 3$

1 point is earned for the reason

$x = 1$ and $x = 3$ because the graph of f' changes from increasing to decreasing at $x = 1$, and changes from decreasing to increasing at $x = 3$.

Part B

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x = 5$ for maximum

1 point is earned for absolute minimum at $x = 4$

1 point is earned for absolute maximum at $x = -1$

The function f decreases from $x = -1$ to $x = 4$, then increases from $x = 4$ to $x = 5$.

Therefore, the absolute minimum value for f is at $x = 4$. The absolute maximum value must occur at $x = -1$ or at $x = 5$.

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x = -1$.

2-8-8



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x = 5$ for maximum

1 point is earned for absolute minimum at $x = 4$

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The function f decreases from $x = -1$ to $x = 4$, then increases from $x = 4$ to $x = 5$.

Therefore, the absolute minimum value for f is at $x = 4$. The absolute maximum value must occur at $x = -1$ or at $x = 5$.

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x = -1$.

Part C

2 points are earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

Tangent line is $y = 4(x - 2) + 12$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

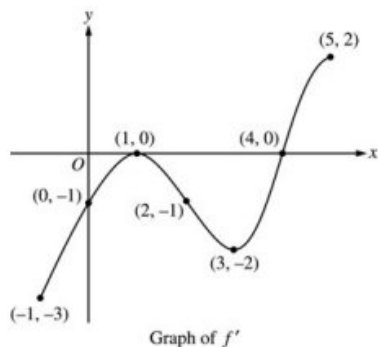
$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

Tangent line is $y = 4(x - 2) + 12$

2-8-8

6.



The figure above shows the graph of f' , the derivative of the function f , on the closed interval $-1 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x=1$ and $x=3$. The function f is twice differentiable with $f'(2)=-6$.

Let g be the function defined by $g(x) = xf(x)$. Find an equation for the line tangent to the graph of g at $x=2$.

Part C

2 point is earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

Tangent line is $y = 4(x-2) + 12$



0	1	2	3
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The student response earns all of the following points:

2 point is earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

Tangent line is $y = 4(x-2) + 12$

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Let $g(x) = xe^{-x} + be^{-x}$, where b is a positive constant.

7. For what positive value of b does g have an absolute maximum at $x = \frac{2}{3}$? Justify your answer.

Part B

2 points are earned for $g'(x)$

1 point is earned for $g'(\frac{2}{3}) = 0$ for b

1 point is earned for justification

$$g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$$

$$g'(\frac{2}{3}) = (\frac{1}{3} - b)e^{2/3} = 0 \Rightarrow b = \frac{1}{3}$$

$$\text{When } b = \frac{1}{3}, g'(x) = (\frac{2}{3} - x)e^{-x}.$$

$$\text{For } x < \frac{2}{3}, g'(x) > 0 \text{ and for } x > \frac{2}{3}, g'(x) < 0.$$

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum

$$\text{at } x = \frac{2}{3}.$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for $g'(\frac{2}{3}) = 0$ for b

1 point is earned for justification

$$g'(x) = e^{-x} - xe^{-x} - be^{-x} = (1 - x - b)e^{-x}$$

$$g'(\frac{2}{3}) = (\frac{1}{3} - b)e^{2/3} = 0 \Rightarrow b = \frac{1}{3}$$

$$\text{When } b = \frac{1}{3}, g'(x) = (\frac{2}{3} - x)e^{-x}.$$

$$\text{For } x < \frac{2}{3}, g'(x) > 0 \text{ and for } x > \frac{2}{3}, g'(x) < 0.$$

Therefore, when $b = \frac{1}{3}$, g has an absolute maximum

$$\text{at } x = \frac{2}{3}.$$

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8. Find all values of b , if any, for which the graph of g has a point of inflection on the interval $0 < x < \infty$. Justify your answer.
-

Part C

2 points are earned for $g''(x)$

1 point is earned for interval for b

1 point is earned for justification

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$. Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for $g''(x)$

1 point is earned for interval for b

1 point is earned for justification

If $0 < b < 2$, then $g''(x)$ will change sign at $x = 2 - b > 0$. Therefore, the graph of g will have a point of inflection on the interval $0 < x < \infty$ when $0 < b < 2$.

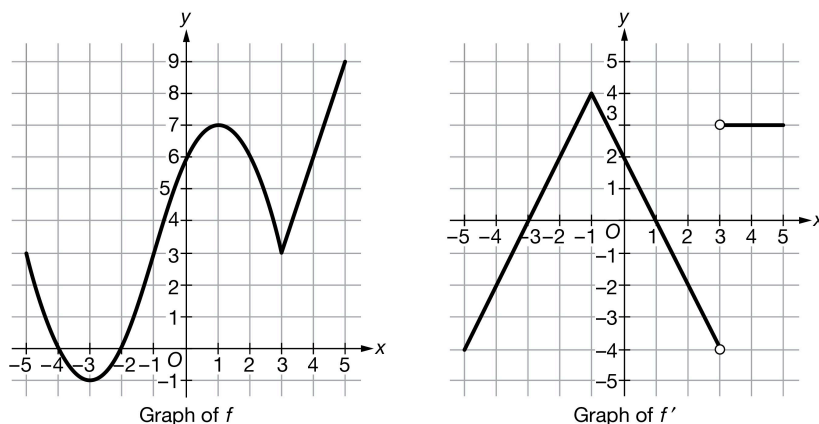
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9. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graphs of the function f and its derivative f' are shown above for $-5 \leq x \leq 5$.

Find the average rate of change of f over the interval $-5 \leq x \leq 5$. For how many values of x in the interval $-5 \leq x \leq 5$ does the instantaneous rate of change of f equal the average rate of change of f over that interval?

Write an equation for the line tangent to the graph of f at $x = 2$.

For each of $\lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)}$ and $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$, find the value or give a reason why it does not exist.

Let g be the function defined by $g(x) = f(x) \cdot \ln x$. Find $g'(4)$.

Part A

If the average rate of change has a maximum of one sign error, the response is eligible for the second point based on a correct interpretation of the graph of f' with the presented average rate of change.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

$$\frac{f(5) - f(-5)}{5 - (-5)} = \frac{9 - 3}{5 - (-5)} = \frac{6}{10} = \frac{3}{5}$$

The average rate of change of f over the interval $-5 \leq x \leq 5$ is $\frac{3}{5}$.

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$f'(x)$, the instantaneous rate of change of f at x , equals $\frac{3}{5}$ for two values of x in the interval $-5 \leq x \leq 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes a correct tangent line equation.

Solution:

$$f(2) = 6$$

$$f'(2) = -2$$

An equation of the line tangent to the graph of f at $x = 2$ is $y = 6 - 2(x - 2)$.

Part C

The second point is earned for “does not exist” with a reason that indicates f is not differentiable at that point (e.g., indicates graph has a corner point or sharp point).

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)}$
- ☐ $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$ does not exist

Solution:

$$\lim_{x \rightarrow -1} \frac{f(x) - f(-1)}{x - (-1)} = f'(-1) = 4$$

$$\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} \text{ does not exist because } f \text{ is not differentiable at } x = 3.$$

Part D

The first point is earned for a correct application of the product rule. The second point is earned for a correct expression that substitutes both correct numerical values.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ answer

Solution:

$$g'(x) = f'(x) \cdot \ln x + f(x) \cdot \frac{1}{x}$$

$$g'(4) = f'(4) \cdot \ln 4 + f(4) \cdot \frac{1}{4} = 3 \ln 4 + 6 \cdot \frac{1}{4} = 3 \ln 4 + \frac{3}{2}$$

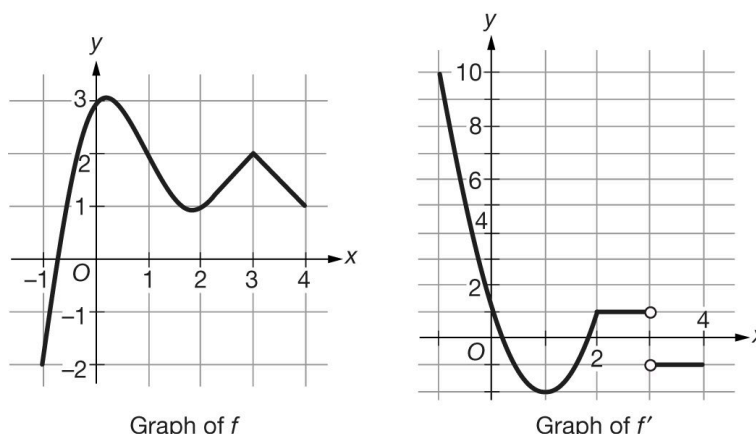
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10. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graphs of the function f and its derivative f' are shown above for $-1 \leq x \leq 4$.

- Find the average rate of change of f over the interval $-1 \leq x \leq 4$. For how many values of x in the interval $-1 \leq x \leq 4$ does the instantaneous rate of change of f equal the average rate of change of f over that interval?
- Write an equation for the line tangent to the graph of f at $x = 1$.
- For each of $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$ and $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$, find the value or give a reason why it does not exist.
- Let g be the function defined by $g(x) = e^x f(x)$. Find $g'(0)$.

Part A

If the average rate of change has a maximum of one sign error, the response is eligible for the second point based on a correct interpretation of the graph of f' with the presented average rate of change.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

$$\frac{f(4) - f(-1)}{4 - (-1)} = \frac{1 - (-2)}{5} = \frac{3}{5}$$

The average rate of change of f over the interval is $\frac{3}{5}$.

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$f'(x)$, the instantaneous rate of change of f at x , equals $\frac{3}{5}$ for two values of x in the interval $[-1, 4]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes a correct tangent line equation.

Solution:

$$f(1) = 2$$

$$f'(1) = -2$$

An equation of the line tangent to the graph of f at $x = 1$ is $y = 2 - 2(x - 1)$.

Part C

The second point is earned for “does not exist” with a reason that indicates f is not differentiable at that point (e.g., indicates graph has a corner point or sharp point).

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$
- ☐ $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$ does not exist

Solution:

$$\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = f'(2) = 1$$

$$\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} \text{ does not exist because } f \text{ is not differentiable at } x = 3.$$

Part D

The first point is earned for a correct application of the product rule. The second point is earned for a correct expression that substitutes both correct numerical values.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2
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The student response accurately includes both of the criteria below.

☐ $g'(x)$

☐ answer

Solution:

$$g'(x) = e^x f(x) + e^x f'(x)$$

$$g'(0) = e^0 f(0) + e^0 f'(0) = 3 + 1 = 4$$

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11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	2	2.2	2.4
$g'(x)$	25	49	81

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(2) = 5$. Selected values of g' are given in the table above. The function g is three-times differentiable.

(a) Write an equation for the line tangent to the graph of g at $x = 2$.

(b) Let h be the function defined by $h(x) = 9 + x \cdot g(x)$. Find $h'(2)$.

(c) Use a right Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_2^{2.4} g'(x) dx$. Show the computations that lead to your answer.

(d) It is known that $g''(x) > 0$ for $2 \leq x \leq 2.4$. Is the approximation from part (c) an overestimate or an underestimate for $\int_2^{2.4} g'(x) dx$? Give a reason for your answer.

(e) Use the approximation for $\int_2^{2.4} g'(x) dx$ from part (c) to estimate the value of $g(2.4)$. Show the computations that lead to your answer.

(f) Use Euler's method, starting at $x = 2$ with two steps of equal size, to approximate $g(2.4)$. Show the computations that lead to your answer.

(g) It is known that $g''(2) = 100$ and $g'''(2) = 200$. Write the third-degree Taylor polynomial for g about $x = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must use the numerical values 5 and 25.

As there is only one point for this part, any simplification errors will not earn the point, even in the presence of a correct answer.

We do not need to see the first line of the model solution. A correct tangent line will earn the point.

Even though we have defined the function $g(x)$ in the stem of the problem, we will accept responses that replace $g(x)$ in place of y in the tangent line equation. For example, $g(x) = 25(x - 2) + 5$ earns the point.

All correct forms of the tangent line earn this point. For example:

$$y - 5 = 25(x - 2)$$

$$y = 5 + 25(x - 2)$$

$$y = 25x - 50 + 5$$

$$y = 25x - 45$$

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0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$y = g(2) + g'(2)(x - 2)$$

$$y = 5 + 25(x - 2)$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Once you see a correct application of the product rule, the response has earned the point. If the response continues and makes an error in simplifying, the response will not earn the second (answer) point.

The first point is for an essentially correct application of the product rule. The point is earned only for a derivative that is one of the following:

$$\cdot h'(x) = g(x) + x g'(x); \text{ or}$$

$$\cdot h'(x) = 9 + g(x) + x g'(x).$$

Note that the mistake with the derivative of the constant in the second example above will make the response ineligible for the second point.

To earn the answer point, we must see a numerical value equivalent to 55. Any arithmetic or algebraic errors in simplification will not earn the answer point, even in the presence of the correct answer.

A response of $5 + 2 \cdot 25$ is the minimal work required to earn both points.

Responses of just $5 + 50$ or 55 would not earn either point because there is no supporting work.

Responses with some values of x plugged in, but not all, may still earn both points. For example, $h'(2) = g(2) + x g'(2) = 55$ will earn both points.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$h'(x) = 1 \cdot g(x) + x \cdot g'(x)$$

$$h'(2) = 1 \cdot g(2) + 2 \cdot g'(2) = 1 \cdot 5 + 2 \cdot 25 = 55$$

Part C

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response must demonstrate the Riemann sum in order to earn this point. This means we must see a sum of two products.

· This could be done with a response of just $49 \cdot 0.2 + 81 \cdot 0.2$ or $0.2(49 + 81)$.

· A response of just $9.8 + 16.2$ does not earn the point.

The answer must be presented numerically. Responses that stop with $g'(2.2) \cdot 0.2 + g'(2.4) \cdot 0.2$ do not earn the point, as we need to see values pulled from the given table.

The answer does not need to be simplified to 26, but if the answer is simplified, this must be done correctly.



0	1
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The student response accurately includes the criteria below.

☐ Right Riemann sum and answer

Solution:

$$\begin{aligned} \int_2^{2.4} g'(x) dx &\approx g'(2.2)(2.2 - 2) + g'(2.4)(2.4 - 2.2) \\ &= 49 \cdot 0.2 + 81 \cdot 0.2 = 26 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a statement that is equivalent to “ g' increasing,” but that is not simply a reference to concavity. This will require describing the behavior of g' or g . For example, it will suffice to say, “the slopes of g are increasing,” to earn the first point. However, saying that “ g is concave up” will not suffice by itself.

A response of just $g'' > 0$ without a conclusion of how this is tied to g' does not earn the first point.

Ambiguous statements will not earn the first point. For example, it is not sufficient to say, “the slopes are increasing,” as it is not clear which function is being referenced. References to “the graph” or “the curve” will also be viewed as ambiguous, as these terms have not been defined in the problem.

Ambiguous use of “the function” will be interpreted as the function g as this is the only “function” defined in the stem. If a response earns the first point and goes on to say something incorrect, the response will not earn the second point.

If the response earns the first point and goes on to say something irrelevant (but not incorrect), the response is still eligible to earn the second point.

Statements whose truth cannot be determined from the information given will also not earn the second point. There are a variety of these that you may see, such as “ g'' is concave up” or “ g' is concave up.”

It is not possible to earn the second point without earning the first point, even in the presence of the correct conclusion of “overestimate.”

Special case: If a response has used $g'' > 0$ to incorrectly conclude that g' is decreasing on the interval, and then answers that the approximation is an underestimate, the response will earn one point.

2-8-8



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ g' increasing
- ☐ Answer

Solution:

Because $g'' > 0$ for $2 \leq x \leq 2.4$, g' is increasing on this interval.

Therefore, the right Riemann sum approximation for $\int_2^{2.4} g'(x) dx$ is an overestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

This point is earned only for a numerical answer. This answer does not need to be simplified, but any attempts at simplification must be correct.

The response must show at least the computation $5 + 26$ to earn this point. An answer of just 31 would not earn the point.

The response will not be penalized for failing to use $\approx$.

Responses are expected to use the approximation of the definite integral found in part (c). A response may do this even with an incorrect approximation found in part (c). A response may import an incorrect answer from part (c) and still earn the point in part (e).

The only acceptable answer in part (e) is the correct answer of 31 or an answer that is consistent with the approximation value found in part (c). (Note that when using an incorrect approximation, the minimal work required would be $5 + (\text{answer from part (c)})$.)



0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\int_2^{2.4} g'(x) dx = g(2.4) - g(2)$$

$$g(2.4) = g(2) + \int_2^{2.4} g'(x) dx \approx 5 + 26 = 31$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler's method, presented in a table, or in the form shown in the model solution. This clear understanding would be demonstrated by no more than one error in the two steps in Euler's method.

2-8-8

Responses that use $g'(2.2)$ in the first step and $g'(2.4)$ in the second step (rather than correctly using the values $g'(2)$ and $g'(2.2)$), will earn the first point as long as there are no additional errors. These responses are ineligible for the second point.

The answer point is earned only by the correct answer.

The answer does not need to be simplified; $5 + 25 \cdot 0.2 + 49 \cdot 0.2$ would earn both points.

An answer of 19.8 with no supporting work will earn no points. Similarly, an answer of only $10 + 9.8$ is not sufficient to earn any points.

A tabular answer with or without labels will earn both points if all values are correct, even if the final answer is found only in the table (and not circled or boxed).



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(2.2) \approx g(2) + g'(2)(2.2 - 2) = 5 + 25 \cdot 0.2 = 10$$

$$g(2.4) \approx g(2.2) + g'(2.2)(2.4 - 2.2) \approx 10 + 49 \cdot 0.2 = 19.8$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The coefficients of the presented polynomial do not need to be simplified. (However, errors in simplification would mean the response does not earn the second point.) For example, if a response presents the second line of the model solution, the response has earned both points.

Only polynomials centered at $x = 2$ are eligible for points in part (g). Polynomials centered about any value other than 2 will earn no points in this part (as only one term could be correct).

Responses with at least two terms correctly centered about $x = 2$, with numerical coefficients will earn the first point. Once earned, this point cannot be lost.

Including terms of degree greater than 3 or adding “+ ...” to the presented Taylor polynomial will not earn the second point.

Responses with terms multiplied out will still be eligible for both points, unless the response boxes the expanded polynomial, indicating that the polynomial about $(x - 2)$ is not correct. For example,

$$5 + 25(x - 2) + \frac{100}{2}(x - 2)^2 + \frac{200}{6}(x - 2)^3 = 25x - 45 + 50(x - 2)^2 + \frac{100}{3}(x - 2)^3 \text{ earns both points, unless the right-hand side of the equation is circled or boxed.}$$

Some responses choose to name the polynomial or equate it with our function $g(x)$. We will accept any naming of this polynomial, including $g(x)$, with one caveat. If the response gives it a name with an argument, the argument may not be a number. For example, responses with names such as $g(x)$ or $T_3(x)$ or T are eligible for both points. However, responses that equate the polynomial with $g(2)$ or $T(2)$ are ineligible for the second point.

Special case: A response with the first line of the model solution, $g(2) + g'(2)(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$, will earn one point. (Note: the response has presented a correct form of the Taylor polynomial for $g(x)$ centered at $x = 2$.) Once this point has been earned, it cannot be lost. Any errors substituting numerical values or simplifying make the response ineligible to earn the second point.

2-8-8



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} &g(2) + g'(2)(x-2) + \frac{g''(2)}{2!}(x-2)^2 + \frac{g'''(2)}{3!}(x-2)^3 \\ &= 5 + 25(x-2) + \frac{100}{2}(x-2)^2 + \frac{200}{6}(x-2)^3 \\ &= 5 + 25(x-2) + 50(x-2)^2 + \frac{100}{3}(x-2)^3 \end{aligned}$$

2-8-8

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	3	3.1	3.2
$g'(x)$	50	35	22

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(3) = 9$. Selected values of g' are given in the table above. The function g is three-times differentiable.

(a) Write an equation for the line tangent to the graph of g at $x = 3$.

(b) Let h be the function defined by $h(x) = 4x \cdot g(x)$. Find $h'(3)$.

(c) Use a left Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_3^{3.2} g'(x) dx$. Show the computations that lead to your answer.

(d) It is known that $g''(x) < 0$ for $3 \leq x \leq 3.2$. Is the approximation from part (c) an overestimate or an underestimate for $\int_3^{3.2} g'(x) dx$? Give a reason for your answer.

(e) Use the approximation for $\int_3^{3.2} g'(x) dx$ from part (c) to estimate the value of $g(3.2)$. Show the computations that lead to your answer.

(f) Use Euler's method, starting at $x = 3$ with two steps of equal size, to approximate $g(3.2)$. Show the computations that lead to your answer.

(g) It is known that $g''(3) = -160$ and $g'''(3) = 200$. Write the third-degree Taylor polynomial for g about $x = 3$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must use the numerical values 9 and 50.

As there is only one point for this part, any simplification errors will not earn the point, even in the presence of a correct answer.

We do not need to see the first line of the model solution. A correct tangent line will earn the point.

Even though we have defined the function $g(x)$ in the stem of the problem, we will accept responses that replace $g(x)$ in place of y in the tangent line equation. For example, $g(x) = 50(x - 3) + 9$ earns the point.

All correct forms of the tangent line earn this point. For example:

$$y - 9 = 50(x - 3)$$

$$y = 9 + 50(x - 3)$$

$$y = 50x - 150 + 9$$

$$y = 50x - 141$$

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$y = g(3) + g'(3)(x - 3)$$

$$y = 9 + 50(x - 3)$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Once you see a correct application of the product rule, the response has earned the point. If the response continues and makes an error in simplifying, the response will not earn the second (answer) point.

The first point is for an essentially correct application of the product rule. The point is earned only for a derivative that is one of the following:

$$\cdot h'(x) = 4g(x) + 4xg'(x)$$

To earn the answer point, we must see a numerical value equivalent to 636. Any arithmetic or algebraic errors in simplification will not earn the answer point, even in the presence of the correct answer.

A response of $4 \cdot 9 + 12 \cdot 50$ is the minimal work required to earn both points.

Responses of just $36 + 600$ or just 636 would not earn either point because there is no supporting work

Responses with some values of x plugged in, but not all, may still earn both points. For example, $h'(3) = 4g(3) + 4xg'(3) = 636$ will earn both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$h'(x) = 4 \cdot g(x) + 4x \cdot g'(x)$$

$$h'(3) = 4 \cdot g(3) + 4 \cdot 3 \cdot g'(3) = 4 \cdot 9 + 12 \cdot 50 = 636$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response must demonstrate the Riemann sum in order to earn this point. This means we must see a sum of two products.

2-8-8

· This could be done with a response of just $50 \cdot 0.1 + 35 \cdot 0.1$ or $0.1(50 + 35)$.

· A response of just $5 + 3.5$ does not earn the point.

The answer must be presented numerically. Responses that stop with $g'(3) \cdot 0.1 + g'(3.1) \cdot 0.1$ do not earn the point, as we need to see values pulled from the given table.

The answer does not need to be simplified to 8.5, but if the answer is simplified, this must be done correctly.



0	1
---	---

The student response accurately includes the criteria below.

☐ Left Riemann sum and answer

Solution:

$$\begin{aligned}\int_3^{3.2} g'(x) dx &\approx g'(3)(3.1 - 3) + g'(3.1)(3.2 - 3.1) \\ &= 50 \cdot 0.1 + 35 \cdot 0.1 = 8.5\end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a statement that is equivalent to “ g' decreasing,” but that is not simply a reference to concavity. This will require describing the behavior of g' or g . For example, it will suffice to say, “the slopes of g are decreasing,” to earn the first point. However, saying that “ g is concave down” will not suffice by itself.

A response of just $g'' < 0$ without a conclusion of how this is tied to g' does not earn the first point.

Ambiguous statements will not earn the first point. For example, it is not sufficient to say, “the slopes are decreasing,” as it is not clear which function is being referenced. References to “the graph” or “the curve” will also be viewed as ambiguous, as these terms have not been defined in the problem.

Ambiguous use of “the function” will be interpreted as the function g as this is the only “function” defined in the stem. If a response earns the first point and goes on to say something incorrect, the response will not earn the second point.

If the response earns the first point and goes on to say something irrelevant (but not incorrect), the response is still eligible to earn the second point.

Statements whose truth cannot be determined from the information given will also not earn the second point. There are a variety of these that you may see, such as “ g'' is concave down” or “ g' is concave down.”

It is not possible to earn the second point without earning the first point, even in the presence of the correct conclusion of “overestimate.”

Special case: If a response has used $g'' < 0$ to incorrectly conclude that g' is increasing on the interval, and then answers that the approximation is an underestimate, the response will earn one point.



0	1	2
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The student response accurately includes **both** of the criteria below.

2-8-8

- ☐ Alt text: g prime decreasing
- ☐ Answer

Solution:

Because $g''(x) < 0$ for $3 \leq x \leq 3.2$, g' is decreasing on this interval.

Therefore, the left Riemann sum approximation for $\int_3^{3.2} g'(x) \, dx$ is an overestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

This point is earned only for a numerical answer. This answer does not need to be simplified, but any attempts at simplification must be correct.

The response must show at least the computation $9 + 8.5$ to earn this point. An answer of just 17.5 would not earn the point.

The response will not be penalized for failing to use $\approx$.

Responses are expected to use the approximation of the definite integral found in part (c). A response may do this even with an incorrect approximation found in part (c). A response may import an incorrect answer from part (c) and still earn the point in part (e).

The only acceptable answer in part (e) is the correct answer of 17.5 or an answer that is consistent with the approximation value found in part (c). (Note that when using an incorrect approximation, the minimal work required would be $9 + (\text{answer from part (c)})$.)



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\int_3^{3.2} g'(x) \, dx = g(3.2) - g(3)$$

$$g(3.2) = g(3) + \int_3^{3.2} g'(x) \, dx \approx 9 + 8.5 = 17.5$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler's method, presented in a table, or in the form shown in the model solution. This clear understanding would be demonstrated by no more than one error in the two steps in Euler's method.

Responses that use $g'(3.1)$ in the first step and $g'(3.2)$ in the second step (rather than correctly using the values $g'(3)$ and $g'(3.1)$), will earn the first point as long as there are no additional errors. These responses are ineligible for the second point.

The answer point is earned only by the correct answer.

The answer does not need to be simplified; $9 + 50 \cdot 0.1 + 35 \cdot 0.1$ would earn both points.

2-8-8

An answer of 17.5 with no supporting work will earn no points. Similarly, an answer of only $14 + 3.5$ is not sufficient to earn any points.

A tabular answer with or without labels will earn both points if all values are correct, even if the final answer is found only in the table (and not circled or boxed).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(3.1) \approx g(3) + g'(3)(3.1 - 3) = 9 + 50 \cdot 0.1 = 14$$

$$g(3.2) \approx g(3.1) + g'(3.1)(3.2 - 3.1) \approx 14 + 35 \cdot 0.1 = 17.5$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The coefficients of the presented polynomial do not need to be simplified. (However, errors in simplification would mean the response does not earn the second point.) For example, if a response presents the second line of the model solution, the response has earned both points.

Only polynomials centered at $x = 3$ are eligible for points in part (g). Polynomials centered about any value other than 3 will earn no points in this part (as only one term could be correct).

Responses with at least two terms correctly centered about $x = 3$, with numerical coefficients will earn the first point. Once earned, this point cannot be lost.

Including terms of degree greater than 3 or adding “+ . . .” to the presented Taylor polynomial will not earn the second point.

Responses with terms multiplied out will still be eligible for both points, unless the response boxes the expanded polynomial, indicating that the polynomial about $(x - 3)$ is not correct. For example,

$9 + 50(x - 3) - \frac{160}{2}(x - 3)^2 + \frac{200}{6}(x - 3)^3 = 50x - 141 - 80(x - 3)^2 + \frac{100}{3}(x - 3)^3$ earns both points, unless the right-hand side of the equation is circled or boxed.

Some responses choose to name the polynomial or equate it with our function $g(x)$. We will accept any naming of this polynomial, including $g(x)$, with one caveat. If the response gives it a name with an argument, the argument may not be a number. For example, responses with names such as $g(x)$ or $T_3(x)$ or T are eligible for both points. However, responses that equate the polynomial with $g(3)$ or $T(3)$ are ineligible for the second point.

Special case: A response with the first line of the model solution, $g(3) + g'(3)(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$, will earn one point. (Note: the response has presented a correct form of the Taylor polynomial for $g(x)$ centered at $x = 3$.) Once this point has been earned, it cannot be lost. Any errors substituting numerical values or simplifying make the response ineligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

2-8-8☐ Two terms☐ Two terms**Solution:**

$$\begin{aligned} g(3) + g'(3)(x-3) + \frac{g''(3)}{2!}(x-3)^2 + \frac{g'''(3)}{3!}(x-3)^3 \\ = 9 + 50(x-3) + \frac{-160}{2}(x-3)^2 + \frac{200}{6}(x-3)^3 \\ = 9 + 50(x-3) - 80(x-3)^2 + \frac{100}{3}(x-3)^3 \end{aligned}$$

2-8-8

13. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	4	4.2	4.4
$g'(x)$	32	50	72

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(4) = 3$. Selected values of g' are given in the table above. The function g is three-times differentiable.

(a) Write an equation for the line tangent to the graph of g at $x = 4$.

(b) Let h be the function defined by $h(x) = 15 + x \cdot g(x)$. Find $h'(4)$.

(c) Use a right Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_4^{4.4} g'(x) dx$. Show the computations that lead to your answer.

(d) It is known that $g''(x) > 0$ for $4 \leq x \leq 4.4$. Is the approximation from part (c) an overestimate or an underestimate for $\int_4^{4.4} g'(x) dx$? Give a reason for your answer.

(e) Use the approximation for $\int_4^{4.4} g'(x) dx$ from part (c) to estimate the value of $g(4.4)$. Show the computations that lead to your answer.

(f) Use Euler's method, starting at $x = 4$ with two steps of equal size, to approximate $g(4.4)$. Show the computations that lead to your answer.

(g) It is known that $g''(4) = 80$ and $g'''(4) = 100$. Write the third-degree Taylor polynomial for g about $x = 4$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must use the numerical values 3 and 32.

As there is only one point for this part, any simplification errors will not earn the point, even in the presence of a correct answer.

We do not need to see the first line of the model solution. A correct tangent line will earn the point.

Even though we have defined the function $g(x)$ in the stem of the problem, we will accept responses that replace $g(x)$ in place of y in the tangent line equation. For example, $g(x) = 32(x - 4) + 3$ earns the point.

All correct forms of the tangent line earn this point. For example:

$$\cdot y - 3 = 32(x - 4)$$

$$\cdot y = 3 + 32(x - 4)$$

$$\cdot y = 32x - 128 + 3$$

$$\cdot y = 32x - 125$$

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$y = g(4) + g'(4)(x - 4)$$

$$y = 3 + 32(x - 4)$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Once you see a correct application of the product rule, the response has earned the point. If the response continues and makes an error in simplifying, the response will not earn the second (answer) point.

The first point is for an essentially correct application of the product rule. The point is earned only for a derivative that is one of the following:

$$\cdot h'(x) = g(x) + x g'(x); \text{ or}$$

$$\cdot h'(x) = 15 + g(x) + x g'(x).$$

Note that the mistake with the derivative of the constant in the second example above will make the response ineligible for the second point.

To earn the answer point, we must see a numerical value equivalent to 131. Any arithmetic or algebraic errors in simplification will not earn the answer point, even in the presence of the correct answer.

A response of $3 + 4 \cdot 32$ is the minimal work required to earn both points.

Responses of just $3 + 128$ or just 131 would not earn either point because there is no supporting work.

Responses with some values of x plugged in, but not all, may still earn both points. For example, $h'(4) = g(4) + x g'(4) = 131$ will earn both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$h'(x) = 1 \cdot g(x) + x \cdot g'(x)$$

$$h'(4) = 1 \cdot g(4) + 4 \cdot g'(4) = 1 \cdot 3 + 4 \cdot 32 = 131$$

Part C

2-8-8

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response must demonstrate the Riemann sum in order to earn this point. This means we must see a sum of two products.

· This could be done with a response of just $50 \cdot 0.2 + 72 \cdot 0.2$ or $0.2(50 + 72)$.

· A response of just $10 + 14.4$ does not earn the point.

The answer must be presented numerically. Responses that stop with $g'(4.2) \cdot 0.2 + g'(4.4) \cdot 0.2$ do not earn the point, as we need to see values pulled from the given table.

The answer does not need to be simplified to 24.4, but if the answer is simplified, this must be done correctly.



0	1
---	---

The student response accurately includes the criteria below.

☐ Right Riemann sum and answer

Solution:

$$\begin{aligned}\int_4^{4.4} g'(x) dx &\approx g'(4.2)(4.2 - 4) + g'(4.4)(4.4 - 4.2) \\ &= 50 \cdot 0.2 + 72 \cdot 0.2 = 24.4\end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a statement that is equivalent to “ g' increasing,” but that is not simply a reference to concavity. This will require describing the behavior of g' or g . For example, it will suffice to say, “the slopes of g are increasing,” to earn the first point. However, saying that “ g is concave up” will not suffice by itself.

A response of just $g'' > 0$ without a conclusion of how this is tied to g' does not earn the first point.

Ambiguous statements will not earn the first point. For example, it is not sufficient to say, “the slopes are increasing,” as it is not clear which function is being referenced. References to “the graph” or “the curve” will also be viewed as ambiguous, as these terms have not been defined in the problem.

Ambiguous use of “the function” will be interpreted as the function g as this is the only “function” defined in the stem. If a response earns the first point and goes on to say something incorrect, the response will not earn the second point.

If the response earns the first point and goes on to say something irrelevant (but not incorrect), the response is still eligible to earn the second point.

Statements whose truth cannot be determined from the information given will also not earn the second point. There are a variety of these that you may see, such as “ g'' is concave up” or “ g' is concave up.”

It is not possible to earn the second point without earning the first point, even in the presence of the correct conclusion of “overestimate.”

Special case: If a response has used $g'' > 0$ to incorrectly conclude that g' is decreasing on the interval, and then answers that the approximation is an underestimate, the response will earn one point.

2-8-8

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ g' increasing
- ☐ Answer

Solution:

Because $g''(x) > 0$ for $4 \leq x \leq 4.4$, g' is increasing on this interval.

Therefore, the right Riemann sum approximation for $\int_4^{4.4} g'(x) \, dx$ is an overestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

This point is earned only for a numerical answer. This answer does not need to be simplified, but any attempts at simplification must be correct.

The response must show at least the computation $3 + 24.4$ to earn this point. An answer of just 27.4 would not earn the point.

The response will not be penalized for failing to use $\approx$.

Responses are expected to use the approximation of the definite integral found in part (c). A response may do this even with an incorrect approximation found in part (c). A response may import an incorrect answer from part (c) and still earn the point in part (e).

The only acceptable answer in part (e) is the correct answer of 27.4 or an answer that is consistent with the approximation value found in part (c). (Note that when using an incorrect approximation, the minimal work required would be $3 + (\text{answer from part (c)})$.)

✓

0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\int_4^{4.4} g'(x) \, dx = g(4.4) - g(4)$$

$$g(4.4) = g(4) + \int_4^{4.4} g'(x) \, dx \approx 3 + 24.4 = 27.4$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler's method, presented in a table, or in the form shown in the model solution. This clear understanding would be demonstrated by no more than one error in the two steps in Euler's method.

2-8-8

Responses that use $g'(4.2)$ in the first step and $g'(4.4)$ in the second step (rather than correctly using the values $g'(4)$ and $g'(4.2)$), will earn the first point as long as there are no additional errors. These responses are ineligible for the second point.

The answer point is earned only by the correct answer.

The answer does not need to be simplified; $3 + 32 \cdot 0.2 + 50 \cdot 0.2$ would earn both points.

An answer of 19.4 with no supporting work will earn no points. Similarly, an answer of only $9.4 + 10$ is not sufficient to earn any points.

A tabular answer with or without labels will earn both points if all values are correct, even if the final answer is found only in the table (and not circled or boxed).



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(4.2) \approx g(4) + g'(4)(4.2 - 4) = 3 + 32 \cdot 0.2 = 9.4$$

$$g(4.4) \approx g(4.2) + g'(4.2)(4.4 - 4.2) \approx 9.4 + 50 \cdot 0.2 = 19.4$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The coefficients of the presented polynomial do not need to be simplified. (However, errors in simplification would mean the response does not earn the second point.) For example, if a response presents the second line of the model solution, the response has earned both points.

Only polynomials centered at $x = 4$ are eligible for points in part (g). Polynomials centered about any value other than 4 will earn no points in this part (as only one term could be correct).

Responses with at least two terms correctly centered about $x = 4$, with numerical coefficients will earn the first point. Once earned, this point cannot be lost.

Including terms of degree greater than 3 or adding “+ ...” to the presented Taylor polynomial will not earn the second point.

Responses with terms multiplied out will still be eligible for both points, unless the response boxes the expanded polynomial, indicating that the polynomial about $(x - 4)$ is not correct. For example,

$3 + 32(x - 4) + \frac{80}{2}(x - 4)^2 + \frac{100}{6}(x - 4)^3 = 32x - 125 + 40(x - 4)^2 + \frac{50}{3}(x - 4)^3$ earns both points, unless the right-hand side of the equation is circled or boxed.

Some responses choose to name the polynomial or equate it with our function $g(x)$. We will accept any naming of this polynomial, including $g(x)$, with one caveat. If the response gives it a name with an argument, the argument may not be a number. For example, responses with names such as $g(x)$ or $T_3(x)$ or T are eligible for both points. However, responses that equate the polynomial with $g(4)$ or $T(4)$ are ineligible for the second point.

Special case: A response with the first line of the model solution, $g(4) + g'(4)(x - 4) + \frac{g''(4)}{2!}(x - 4)^2 + \frac{g'''(4)}{3!}(x - 4)^3$, will earn one point. (Note: the response has presented a correct form of the Taylor polynomial for $g(x)$ centered at $x = 4$.) Once this point has been earned, it cannot be lost. Any errors substituting numerical values or simplifying make the response ineligible to earn the second point.

2-8-8



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} & g(4) + g'(4)(x-4) + \frac{g''(4)}{2!}(x-4)^2 + \frac{g'''(4)}{3!}(x-4)^3 \\ &= 3 + 32(x-4) + \frac{80}{2}(x-4)^2 + \frac{100}{6}(x-4)^3 \\ &= 3 + 32(x-4) + 40(x-4)^2 + \frac{50}{3}(x-4)^3 \end{aligned}$$

2-8-8

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	3	3.2	3.4
$g'(x)$	16	30	48

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(3) = 7$. Selected values of g' are given in the table above. The function g is three-times differentiable.

(a) Write an equation for the line tangent to the graph of g at $x = 3$.

(b) Let h be the function defined by $h(x) = -5 + x \cdot g(x)$. Find $h'(3)$.

(c) Use a left Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_3^{3.4} g'(x) dx$. Show the computations that lead to your answer.

(d) It is known that $g''(x) > 0$ for $3 \leq x \leq 3.4$. Is the approximation from part (c) an overestimate or an underestimate for $\int_3^{3.4} g'(x) dx$? Give a reason for your answer.

(e) Use the approximation for $\int_3^{3.4} g'(x) dx$ from part (c) to estimate the value of $g(3.4)$. Show the computations that lead to your answer.

(f) Use Euler's method, starting at $x = 3$ with two steps of equal size, to approximate $g(3.4)$. Show the computations that lead to your answer.

(g) It is known that $g''(3) = 60$ and $g'''(3) = 100$. Write the third-degree Taylor polynomial for g about $x = 3$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must use the numerical values 7 and 16.

As there is only one point for this part, any simplification errors will not earn the point, even in the presence of a correct answer.

We do not need to see the first line of the model solution. A correct tangent line will earn the point.

Even though we have defined the function $g(x)$ in the stem of the problem, we will accept responses that replace $g(x)$ in place of y in the tangent line equation. For example, $g(x) = 16(x - 3) + 7$ earns the point.

All correct forms of the tangent line earn this point. For example:

$$y - 7 = 16(x - 3)$$

$$y = 7 + 16(x - 3)$$

$$y = 16x - 48 + 7$$

$$y = 16x - 41$$

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$y = g(3) + g'(3)(x - 3)$$

$$y = 7 + 16(x - 3)$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Once you see a correct application of the product rule, the response has earned the point. If the response continues and makes an error in simplifying, the response will not earn the second (answer) point.

The first point is for an essentially correct application of the product rule. The point is earned only for a derivative that is one of the following:

$$\cdot h'(x) = g(x) + x g'(x); \text{ or}$$

$$\cdot h'(x) = -5 + g(x) + x g'(x).$$

Note that the mistake with the derivative of the constant in the second example above will make the response ineligible for the second point.

To earn the answer point, we must see a numerical value equivalent to 55. Any arithmetic or algebraic errors in simplification will not earn the answer point, even in the presence of the correct answer.

A response of $7 + 3 \cdot 16$ is the minimal work required to earn both points.

Responses of just $7 + 48$ or just 55 would not earn either point because there is no supporting work.

Responses with some values of x plugged in, but not all, may still earn both points. For example, $h'(3) = g(3) + x g'(3) = 55$ will earn both points.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$h'(x) = 1 \cdot g(x) + x \cdot g'(x)$$

$$h'(3) = 1 \cdot g(3) + 3 \cdot g'(3) = 1 \cdot 7 + 3 \cdot 16 = 55$$

Part C

2-8-8

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response must demonstrate the Riemann sum in order to earn this point. This means we must see a sum of two products.

· This could be done with a response of just $16 \cdot 0.2 + 30 \cdot 0.2$ or $0.2(16 + 30)$.

· A response of just $3.2 + 6$ does not earn the point.

The answer must be presented numerically. Responses that stop with $g'(3) \cdot 0.2 + g'(3.2) \cdot 0.2$ do not earn the point, as we need to see values pulled from the given table.

The answer does not need to be simplified to 9.2, but if the answer is simplified, this must be done correctly.



0	1
---	---

The student response accurately includes the criteria below.

☐ Left Riemann sum and answer

Solution:

$$\begin{aligned} \int_3^{3.4} g'(x) dx &\approx g'(3)(3.2 - 3) + g'(3.2)(3.4 - 3.2) \\ &= 16 \cdot 0.2 + 30 \cdot 0.2 = 9.2 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a statement that is equivalent to “ g' increasing,” but that is not simply a reference to concavity. This will require describing the behavior of g' or g . For example, it will suffice to say, “the slopes of g are increasing,” to earn the first point. However, saying that “ g is concave up” will not suffice by itself.

A response of just $g'' > 0$ is greater than 0 without a conclusion of how this is tied to g' does not earn the first point.

Ambiguous statements will not earn the first point. For example, it is not sufficient to say, “the slopes are increasing,” as it is not clear which function is being referenced. References to “the graph” or “the curve” will also be viewed as ambiguous, as these terms have not been defined in the problem.

Ambiguous use of “the function” will be interpreted as the function g as this is the only “function” defined in the stem. If a response earns the first point and goes on to say something incorrect, the response will not earn the second point.

If the response earns the first point and goes on to say something irrelevant (but not incorrect), the response is still eligible to earn the second point.

Statements whose truth cannot be determined from the information given will also not earn the second point. There are a variety of these that you may see, such as “ g'' is concave up” or “ g' is concave up.”

It is not possible to earn the second point without earning the first point, even in the presence of the correct conclusion of “underestimate.”

Special case: If a response has used $g'' > 0$ to incorrectly conclude that g' is decreasing on the interval, and then answers that the approximation is an overestimate, the response will earn one point.

2-8-8

✓

0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ g' increasing
- ☐ Answer

Solution:

Because $g''(x) > 0$ for $3 \leq x \leq 3.4$, g' is increasing on this interval.

Therefore, the left Riemann sum approximation for $\int_3^{3.4} g'(x) \, dx$ is an underestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

This point is earned only for a numerical answer. This answer does not need to be simplified, but any attempts at simplification must be correct.

The response must show at least the computation $7 + 9.2$ to earn this point. An answer of just 16.2 would not earn the point.

The response will not be penalized for failing to use $\approx$.

Responses are expected to use the approximation of the definite integral found in part (c). A response may do this even with an incorrect approximation found in part (c). A response may import an incorrect answer from part (c) and still earn the point in part (e).

The only acceptable answer in part (e) is the correct answer of 16.2 or an answer that is consistent with the approximation value found in part (c). (Note that when using an incorrect approximation, the minimal work required would be $7 + (\text{answer from part (c)})$.)

✓

0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\int_3^{3.4} g'(x) \, dx = g(3.4) - g(3)$$

$$g(3.4) = g(3) + \int_3^{3.4} g'(x) \, dx \approx 7 + 9.2 = 16.2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler's method, presented in a table, or in the form shown in the model solution. This clear understanding would be demonstrated by no more than one error in the two steps in Euler's method.

2-8-8

Responses that use $g'(3.2)$ in the first step and $g'(3.4)$ in the second step (rather than correctly using the values $g'(3)$ and $g'(3.2)$), will earn the first point as long as there are no additional errors. These responses are ineligible for the second point.

The answer point is earned only by the correct answer.

The answer does not need to be simplified; $7 + 16 \cdot 0.2 + 30 \cdot 0.2$ would earn both points.

An answer of 16.2 with no supporting work will earn no points. Similarly, an answer of only $10.2 + 6$ is not sufficient to earn any points.

A tabular answer with or without labels will earn both points if all values are correct, even if the final answer is found only in the table (and not circled or boxed).



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(3.2) \approx g(3) + g'(3)(3.2 - 3) = 7 + 16 \cdot 0.2 = 10.2$$

$$g(3.4) \approx g(3.2) + g'(3.2)(3.4 - 3.2) \approx 10.2 + 30 \cdot 0.2 = 16.2$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The coefficients of the presented polynomial do not need to be simplified. (However, errors in simplification would mean the response does not earn the second point.) For example, if a response presents the second line of the model solution, the response has earned both points.

Only polynomials centered at $x = 3$ are eligible for points in part (g). Polynomials centered about any value other than 3 will earn no points in this part (as only one term could be correct).

Responses with at least two terms correctly centered about $x = 3$, with numerical coefficients will earn the first point. Once earned, this point cannot be lost.

Including terms of degree greater than 3 or adding “+ ...” to the presented Taylor polynomial will not earn the second point.

Responses with terms multiplied out will still be eligible for both points, unless the response boxes the expanded polynomial, indicating that the polynomial about $(x - 3)$ is not correct. For example,

$7 + 16(x - 3) + \frac{60}{2}(x - 3)^2 + \frac{100}{6}(x - 3)^3 = 16x - 41 + 30(x - 3)^2 + \frac{50}{3}(x - 3)^3$ earns both points, unless the right-hand side of the equation is circled or boxed.

Some responses choose to name the polynomial or equate it with our function $g(x)$. We will accept any naming of this polynomial, including $g(x)$, with one caveat. If the response gives it a name with an argument, the argument may not be a number. For example, responses with names such as $g(x)$ or $T_3(x)$ or T are eligible for both points. However, responses that equate the polynomial with $g(3)$ or $T(3)$ are ineligible for the second point.

Special case: A response with the first line of the model solution, $g(3) + g'(3)(x - 3) + \frac{g''(3)}{2!}(x - 3)^2 + \frac{g'''(3)}{3!}(x - 3)^3$, will earn one point. (Note: the response has presented a correct form of the Taylor polynomial for $g(x)$ centered at $x = 3$.) Once this point has been earned, it cannot be lost. Any errors substituting numerical values or simplifying make the response ineligible to earn the second point.

2-8-8



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} & g(3) + g'(3)(x-3) + \frac{g''(3)}{2!}(x-3)^2 + \frac{g'''(3)}{3!}(x-3)^3 \\ &= 7 + 16(x-3) + \frac{60}{2}(x-3)^2 + \frac{100}{6}(x-3)^3 \\ &= 7 + 16(x-3) + 30(x-3)^2 + \frac{50}{3}(x-3)^3 \end{aligned}$$

2-8-8

15. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	2	2.1	2.2
$g'(x)$	64	41	20

Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = g'(x)$ with initial condition $g(2) = 7$. Selected values of g' are given in the table above. The function g is three-times differentiable.

(a) Write an equation for the line tangent to the graph of g at $x = 2$.

(b) Let h be the function defined by $h(x) = 3x \cdot g(x)$. Find $h'(2)$.

(c) Use a right Riemann sum with the two subintervals indicated by the data in the table to approximate $\int_2^{2.2} g'(x) dx$. Show the computations that lead to your answer.

(d) It is known that $g''(x) < 0$ for $2 \leq x \leq 2.2$. Is the approximation from part (c) an overestimate or an underestimate for $\int_2^{2.2} g'(x) dx$? Give a reason for your answer.

(e) Use the approximation for $\int_2^{2.2} g'(x) dx$ from part (c) to estimate the value of $g(2.2)$. Show the computations that lead to your answer.

(f) Use Euler's method, starting at $x = 2$ with two steps of equal size, to approximate $g(2.2)$. Show the computations that lead to your answer.

(g) It is known that $g''(2) = -240$ and $g'''(2) = 200$. Write the third-degree Taylor polynomial for g about $x = 2$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must use the numerical values 7 and 64.

As there is only one point for this part, any simplification errors will not earn the point, even in the presence of a correct answer.

We do not need to see the first line of the model solution. A correct tangent line will earn the point.

Even though we have defined the function $g(x)$ in the stem of the problem, we will accept responses that replace $g(x)$ in place of y in the tangent line equation. For example, $g(x) = 64(x - 2) + 7$ earns the point.

All correct forms of the tangent line earn this point. For example:

$$y - 7 = 64(x - 2)$$

$$y = 7 + 64(x - 2)$$

$$y = 64x - 128 + 7$$

$$y = 64x - 121$$

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$y = g(2) + g'(2)(x - 2)$$

$$y = 7 + 64(x - 2)$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Once you see a correct application of the product rule, the response has earned the point. If the response continues and makes an error in simplifying, the response will not earn the second (answer) point.

The first point is for an essentially correct application of the product rule. The point is earned only for a derivative equivalent to the following:

$$h'(x) = 3g(x) + 3xg'(x)$$

To earn the answer point, we must see a numerical value equivalent to 405. Any arithmetic or algebraic errors in simplification will not earn the answer point, even in the presence of the correct answer.

A response of $3 \cdot 7 + 6 \cdot 64$ is the minimal work required to earn both points.

Responses of just $21 + 384$ or just 405 would not earn either point because there is no supporting work.

Responses with some values of x plugged in, but not all, may still earn both points. For example, $h'(2) = 3g(2) + 3xg'(2) = 405$ will earn both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$h'(x) = 3 \cdot g(x) + 3x \cdot g'(x)$$

$$h'(2) = 3 \cdot g(2) + 3 \cdot 2 \cdot g'(2) = 3 \cdot 7 + 6 \cdot 64 = 405$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response must demonstrate the Riemann sum in order to earn this point. This means we must see a sum of two products.

2-8-8

· This could be done with a response of just $41 \cdot 0.1 + 20 \cdot 0.1$ or $0.1(41 + 20)$.

· A response of just $4.1 + 2$ does not earn the point.

The answer must be presented numerically. Responses that stop with $g'(2.1) \cdot 0.1 + g'(2.2) \cdot 0.1$ do not earn the point, as we need to see values pulled from the given table.

The answer does not need to be simplified to 6.1, but if the answer is simplified, this must be done correctly.



0	1
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The student response accurately includes the criteria below.

☐ Right Riemann sum and answer

Solution:

$$\begin{aligned} \int_2^{2.2} g'(x) dx &\approx g'(2.1)(2.1 - 2) + g'(2.2)(2.2 - 2.1) \\ &= 41 \cdot 0.1 + 20 \cdot 0.1 = 6.1 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point requires a statement that is equivalent to “ g' decreasing,” but that is not simply a reference to concavity. This will require describing the behavior of g' or g . For example, it will suffice to say, “the slopes of g are decreasing,” to earn the first point. However, saying that “ g is concave down” will not suffice by itself.

A response of just $g'' < 0$ without a conclusion of how this is tied to g' does not earn the first point.

Ambiguous statements will not earn the first point. For example, it is not sufficient to say, “the slopes are decreasing,” as it is not clear which function is being referenced. References to “the graph” or “the curve” will also be viewed as ambiguous, as these terms have not been defined in the problem.

Ambiguous use of “the function” will be interpreted as the function g as this is the only “function” defined in the stem. If a response earns the first point and goes on to say something incorrect, the response will not earn the second point.

If the response earns the first point and goes on to say something irrelevant (but not incorrect), the response is still eligible to earn the second point.

Statements whose truth cannot be determined from the information given will also not earn the second point. There are a variety of these that you may see, such as “ g'' is concave down” or “ g' is concave down.”

It is not possible to earn the second point without earning the first point, even in the presence of the correct conclusion of “underestimate.”

Special case: If a response has used $g'' < 0$ to incorrectly conclude that g' is increasing on the interval, and then answers that the approximation is an overestimate, the response will earn one point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

2-8-8

- ☐ g' decreasing
- ☐ Answer

Solution:

Because $g''(x) < 0$ for $2 \leq x \leq 2.2$, g' is decreasing on this interval.

Therefore, the right Riemann sum approximation for $\int_2^{2.2} g'(x) \, dx$ is an underestimate.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

This point is earned only for a numerical answer. This answer does not need to be simplified, but any attempts at simplification must be correct.

The response must show at least the computation $7 + 6.1$ to earn this point. An answer of just 13.1 would not earn the point.

The response will not be penalized for failing to use $\approx$.

Responses are expected to use the approximation of the definite integral found in part (c). A response may do this even with an incorrect approximation found in part (c). A response may import an incorrect answer from part (c) and still earn the point in part (e).

The only acceptable answer in part (e) is the correct answer of 13.1 or an answer that is consistent with the approximation value found in part (c). (Note that when using an incorrect approximation, the minimal work required would be $7 + (\text{answer from part (c)})$.)



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\int_2^{2.2} g'(x) \, dx = g(2.2) - g(2)$$

$$g(2.2) = g(2) + \int_2^{2.2} g'(x) \, dx \approx 7 + 6.1 = 13.1$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is awarded for a response that provides clear evidence of an understanding of Euler's method, presented in a table, or in the form shown in the model solution. This clear understanding would be demonstrated by no more than one error in the two steps in Euler's method.

Responses that use $g'(2.1)$ in the first step and $g'(2.2)$ in the second step (rather than correctly using the values $g'(2)$ and $g'(2.1)$), will earn the first point as long as there are no additional errors. These responses are ineligible for the second point.

The answer point is earned only by the correct answer.

The answer does not need to be simplified; $7 + 64 \cdot 0.1 + 41 \cdot 0.1$ would earn both points.

2-8-8

An answer of 17.5 with no supporting work will earn no points. Similarly, an answer of only $13.4 + 4.1$ is not sufficient to earn any points.

A tabular answer with or without labels will earn both points if all values are correct, even if the final answer is found only in the table (and not circled or boxed).



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ First step of Euler's method
- ☐ Answer

Solution:

$$g(2.1) \approx g(2) + g'(2)(2.1 - 2) = 7 + 64 \cdot 0.1 = 13.4$$

$$g(2.2) \approx g(2.1) + g'(2.1)(2.2 - 2.1) \approx 13.4 + 41 \cdot 0.1 = 17.5$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The coefficients of the presented polynomial do not need to be simplified. (However, errors in simplification would mean the response does not earn the second point.) For example, if a response presents the second line of the model solution, the response has earned both points.

Only polynomials centered at $x = 2$ are eligible for points in part (g). Polynomials centered about any value other than 2 will earn no points in this part (as only one term could be correct).

Responses with at least two terms correctly centered about $x = 2$, with numerical coefficients will earn the first point. Once earned, this point cannot be lost.

Including terms of degree greater than 3 or adding “+ ...” to the presented Taylor polynomial will not earn the second point.

Responses with terms multiplied out will still be eligible for both points, unless the response boxes the expanded polynomial, indicating that the polynomial about $(x - 2)$ is not correct. For example,

$7 + 64(x - 2) - \frac{240}{2}(x - 2)^2 + \frac{200}{6}(x - 2)^3 = 64x - 121 - 120(x - 2)^2 + \frac{100}{3}(x - 2)^3$ earns both points, unless the right-hand side of the equation is circled or boxed.

Some responses choose to name the polynomial or equate it with our function $g(x)$. We will accept any naming of this polynomial, including $g(x)$, with one caveat. If the response gives it a name with an argument, the argument may not be a number. For example, responses with names such as $g(x)$ or $T_3(x)$ or T are eligible for both points. However, responses that equate the polynomial with $g(2)$ or $T(2)$ are ineligible for the second point.

Special case: A response with the first line of the model solution, $g(2) + g'(2)(x - 2) + \frac{g''(2)}{2!}(x - 2)^2 + \frac{g'''(2)}{3!}(x - 2)^3$, will earn one point. (Note: the response has presented a correct form of the Taylor polynomial for $g(x)$ centered at $x = 2$.) Once this point has been earned, it cannot be lost. Any errors substituting numerical values or simplifying make the response ineligible to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

2-8-8

- ☐ Two terms
- ☐ Remaining terms

Solution:

$$\begin{aligned} g(2) + g'(2)(x-2) + \frac{g''(2)}{2!}(x-2)^2 + \frac{g'''(2)}{3!}(x-2)^3 \\ = 7 + 64(x-2) + \frac{-240}{2}(x-2)^2 + \frac{200}{6}(x-2)^3 \\ = 7 + 64(x-2) - 120(x-2)^2 + \frac{100}{3}(x-2)^3 \end{aligned}$$

2-8-8

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Functions f , g , and h are twice-differentiable functions with $g(2) = h(2) = 4$. The line $y = 4 + \frac{2}{3}(x - 2)$ is tangent to both the graph of g at $x = 2$ and the graph of h at $x = 2$.

a) Find $h'(2)$.

b) Let a be the function given by $a(x) = 3x^3h(x)$. Write an expression for $a'(x)$. Find $a'(2)$.

c) The function h satisfies $h(x) = \frac{x^2 - 4}{1 - (f(x))^3}$ for $x \neq 2$. It is known that $\lim_{x \rightarrow 2} h(x)$ can be evaluated using L'Hospital's Rule. Use $\lim_{x \rightarrow 2} h(x)$ to find $f(2)$ and $f'(2)$. Show the work that leads to your answers.

d) It is known that $g(x) \leq h(x)$ for $1 < x < 3$. Let k be a function satisfying $g(x) \leq k(x) \leq h(x)$ for $1 < x < 3$. Is k continuous at $x = 2$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes the criteria below.

☐ answer

Solution:

$$h'(2) = \frac{2}{3}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for a derivative in the form of the product rule, namely $a'(x) = p(x) \cdot h(x) + q(x) \cdot h'(x)$, where $p(x)$ and $q(x)$ are both polynomials.

Simplification is not required to earn the third point. Function values must be substituted into the expression.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ form of product rule

2-8-8

☐ $a'(x)$

☐ $a'(2)$

Solution:

$$a'(x) = 9x^2h(x) + 3x^3h'(x)$$

$$a'(2) = 9 \cdot 2^2h(2) + 3 \cdot 2^3h'(2) = 36 \cdot 4 + 24 \cdot \frac{2}{3} = 160$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for showing $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$. This work may appear anywhere in the response.

The second point is earned for calculating $f(2) = 0$ in the context of setting the denominator $1 - (f(x))^3$ equal to 0.

The third point is earned for evidence of applying L'Hospital's Rule, which includes a new limit of a quotient where differentiation was attempted in both the numerator and denominator. Note that incorrect notation such as $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = \frac{0}{0}$ makes a response not eligible to earn the third point.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

☐ $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$

☐ $f(2)$

☐ L'Hospital's Rule

☐ $f'(2)$

Solution:

Because h is differentiable, h is continuous, so $\lim_{x \rightarrow 2} h(x) = h(2) = 4$.

Also, $\lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3}$, so $\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = 4$.

Because $\lim_{x \rightarrow 2} (x^2 - 4) = 0$, we must also have $\lim_{x \rightarrow 2} (1 - (f(x))^3) = 0$.

Thus $\lim_{x \rightarrow 2} f(x) = 1$.

Because f is differentiable, f is continuous, so $f(2) = \lim_{x \rightarrow 2} f(x) = 1$.

Also, because f is twice differentiable, f' is continuous, so $\lim_{x \rightarrow 2} f'(x) = f'(2)$ exists.

Using L'Hospital's Rule,

$$\lim_{x \rightarrow 2} \frac{x^2-4}{1-(f(x))^3} = \lim_{x \rightarrow 2} \frac{2x}{-3(f(x))^2 f'(x)} = \frac{4}{-3(1)^2 \cdot f'(2)} = 4.$$

2-8-8

Thus $f'(2) = -\frac{1}{3}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point does not require naming the squeeze theorem. The justification requires (1) stating g and h are continuous, (2) stating $\lim_{x \rightarrow 2} g(x) = 4$ and $\lim_{x \rightarrow 2} h(x) = 4$ and concluding that $\lim_{x \rightarrow 2} k(x) = 4$, and (3) stating $g(2) = 4 = h(2)$ and concluding $k(2) = 4$.



0	1
---	---

The student response accurately includes the criteria below.

☐

continuous with justification

Solution:

Because g and h are differentiable, g and h are continuous, so $\lim_{x \rightarrow 2} g(x) = g(2) = 4$ and $\lim_{x \rightarrow 2} h(x) = h(2) = 4$.

Because $g(x) \leq k(x) \leq h(x)$ for $1 < x < 3$, it follows from the squeeze theorem that $\lim_{x \rightarrow 2} k(x) = 4$.

Also, $4 = g(2) \leq k(2) \leq h(2) = 4$, so $k(2) = 4$.

Thus k is continuous at $x = 2$.

2-8-8

17. Consider the curve defined by $x^2 + xy + y^2 = 27$.

(a) Write an expression for the slope of the curve at any point (x, y) .

(b) Determine whether the lines tangent to the curve at the x -intercepts of the curve are parallel. Show the analysis that leads to your conclusion.

(c) Find the points on the curve where the lines tangent to the curve are vertical.

Part A

2 points are earned for the correct implicit differentiation

<-1> each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct implicit differentiation

<-1> each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$

Part B

1 point is earned for correctly using $y = 0$

$$\text{If } y = 0, x^2 = 27$$

$$x = \pm 3\sqrt{3}$$

1 point is earned for correctly finding y' at x -intercepts

$$\text{at } x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$$

2-8-8

$$\text{at } x = -3\sqrt{3}, y' = \frac{2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$$

1 point is earned for the correct analysis and conclusion

Tangent lines at x -intercepts are parallel

NOTE: For solution at y -intercepts, eligible for 2nd and 3rd points



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $y = 0$

$$\text{If } y = 0, x^2 = 27$$

$$x = \pm 3\sqrt{3}$$

1 point is earned for correctly finding y' at x -intercepts

$$\text{at } x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$$

$$\text{at } x = -3\sqrt{3}, y' = \frac{2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$$

1 point is earned for the correct analysis and conclusion

Tangent lines at x -intercepts are parallel

NOTE: For solution at y -intercepts, eligible for 2nd and 3rd points

Part C

1 point is earned for correctly setting $x + 2y = 0$

y' undefined if $x + 2y = 0$

1 point is earned for correctly solving for y (or x) values

$$\begin{aligned} (-2y)^2 + (-2y)y + y^2 &= 27 \\ 3y^2 &= 27 \\ y &= \pm 3 \end{aligned}$$

1 point is earned for correctly giving both points

Points are $(-6, 3)$ and $(6, -3)$

NOTE: max 1/3 for horizontal tangents

2-8-8

Max 1/3 if denominator of y' is not of form $ax + by + c$ with $a \cdot b \neq 0$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly setting $x + 2y = 0$

y' undefined if $x + 2y = 0$

1 point is earned for correctly solving for y (or x) values

$$\begin{aligned} (-2y)^2 + (-2y)y + y^2 &= 27 \\ 3y^2 &= 27 \\ y &= \pm 3 \end{aligned}$$

1 point is earned for correctly giving both points

Points are $(-6, 3)$ and $(6, -3)$

NOTE: max 1/3 for horizontal tangents

Max 1/3 if denominator of y' is not of form $ax + by + c$ with $a \cdot b \neq 0$.

2-8-8

18. Consider the curve defined by $x^2 + xy + y^2 = 27$.

Write an expression for the slope of the curve at any point (x, y) .

Part A

2 points are earned for the correct implicit differentiation

< -1 > each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct implicit differentiation

< -1 > each incorrect derivative of a term

$$2x + xy' + y + 2yy' = 0$$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$

2-8-8

19. Consider the curve defined by $-8x^2 + 5xy + y^3 = -149$.

(a) Find $\frac{dy}{dx}$.

(b) Write an equation for the line tangent to the curve at the point $(4, -1)$.

(c) There is a number k so that the point $(4.2, k)$ is on the curve. Using the tangent line found in part (b), approximate the value of k .

(d) Write an equation that can be solved to find the actual value of k so that the point $(4.2, k)$ is on the curve.

(e) Solve the equation found in part (d) for the value of k .

Part A

2 points are earned for the correct implicit differentiation

<-1> each error

$$-16x + 5y + 5x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x - 5y}{5x + 3y^2}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct implicit differentiation

<-1> each error

$$-16x + 5y + 5x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x - 5y}{5x + 3y^2}$$

Part B

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

2-8-8

$$\left. \frac{dy}{dx} \right|_{(4,-1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$$y + 1 = 3(x - 4)$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left. \frac{dy}{dx} \right|_{(4,-1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$$y + 1 = 3(x - 4)$$

Part C

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$$y + 1 = 3(4.2 - 4)$$

1 point is earned for the correct solution

<-1> if " $k =$ "

$$y = -0.4$$

$$k \approx -0.4$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$$y + 1 = 3(4.2 - 4)$$

1 point is earned for the correct solution

<-1> if " $k =$ "

$$y = -0.4$$

$$k \approx -0.4$$

Part D

1 point is earned for correctly using $x = 4.2$ in equation of curve

2-8-8

$$-8(4.2)^2 + 5(4.2)k + k^3 = -149$$

✓

0	1
---	---

The student response earns all of the following points:

1 point is earned for correctly using $x = 4.2$ in equation of curve

$$-8(4.2)^2 + 5(4.2)k + k^3 = -149$$

Part E

1 point is earned for the correct answer

$$k = -0.373$$

✓

0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct answer

$$k = -0.373$$

2-8-8

20. Consider the curve defined by $-8x^2 + 5xy + y^3 = -149$.

Find $\frac{dy}{dx}$.

Part A

2 points are earned for the correct implicit differentiation

< -1 > each error

$$-16x + 5y + 5x\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x-5y}{5x+3y^2}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct implicit differentiation

< -1 > each error

$$-16x + 5y + 5x\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x-5y}{5x+3y^2}$$

2-8-8

21. Consider the function f given by $f(x) = xe^{-x^2}$ for all real numbers x .

At what value of x does $f(x)$ attain its absolute maximum? Justify your answer.

Part A

Two points are earned for $f'(x)$

One point is earned for $f'(x) = 0$

One point is earned for the answer

One point is earned for the justification

$$f'(x) = e^{-x^2} - 2x^2e^{-x^2} = (1 - 2x^2)e^{-x^2}$$

$$f'(x) = 0 \text{ when } x = -\frac{1}{\sqrt{2}} \text{ or } x = \frac{1}{\sqrt{2}}$$

$f(x) < 0$ for $x < 0$, and $f(x) > 0$ for $x > 0$. Therefore, a maximum can only occur at a positive x -value. $f'(x)$ changes sign from

positive to negative at $x = \frac{1}{\sqrt{2}}$, so f has a relative maximum at $x = \frac{1}{\sqrt{2}}$. f increases on $0 < x \leq \frac{1}{\sqrt{2}}$ and decreases on $\frac{1}{\sqrt{2}} \leq x < \infty$.

Therefore, there is an absolute maximum at $x = \frac{1}{\sqrt{2}}$.



0	1	2	3	4	5
---	---	---	---	---	---

The student earns all of the following points:

Two points are earned for $f'(x)$

One point is earned for $f'(x) = 0$

One point is earned for the answer

One point is earned for the justification

$$f'(x) = e^{-x^2} - 2x^2e^{-x^2} = (1 - 2x^2)e^{-x^2}$$

$$f'(x) = 0 \text{ when } x = -\frac{1}{\sqrt{2}} \text{ or } x = \frac{1}{\sqrt{2}}$$

2-8-8

$f(x) < 0$ for $x < 0$, and $f(x) > 0$ for $x > 0$. Therefore, a maximum can only occur at a positive x -value. $f'(x)$ changes sign from

positive to negative at $x = \frac{1}{\sqrt{2}}$, so f has a relative maximum at $x = \frac{1}{\sqrt{2}}$. f increases on $0 < x < \frac{1}{\sqrt{2}}$ and decreases on $\frac{1}{\sqrt{2}} \leq x < \infty$.

Therefore, there is an absolute maximum at $x = \frac{1}{\sqrt{2}}$.

22. A particle moves along the y -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = t \cos t$. At time $t = 0$, the position of the particle is $y = 3$.

Write an expression for the acceleration of the particle in terms of t .

Part B

1 point is earned for correctly indicating $a(t) = v'(t)$

1 point is earned for the correct derivative

$$a(t) = v'(t) = \cos t - t \sin t$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly indicating $a(t) = v'(t)$

1 point is earned for the correct derivative

$$a(t) = v'(t) = \cos t - t \sin t$$

2-8-8

23. Let f and g be the functions given by $f(x) = e^x$ and $g(x) = \ln x$.

 Let h be the function given by $h(x) = f(x) - g(x)$. Find the absolute minimum value of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$, and find the absolute maximum of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$. Show the analysis that leads to your answers.

Part C

1 point is earned for considers $h'(x) = 0$

1 point is earned for identifies critical point and endpoints as candidates

1 point is earned for the answers

$$h'(x) = f'(x) - g'(x) = e^x - \frac{1}{x} = 0$$

$$x = 0.567143$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.34418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considers $h'(x) = 0$

1 point is earned for identifies critical point and endpoints as candidates

1 point is earned for the answers

$$h'(x) = f'(x) - g'(x) = e^x - \frac{1}{x} = 0$$

$$x = 0.567143$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.34418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.

2-8-8

24. Let f be the function given by $f(x) = (2x - 1)^5(x + 1)$. Which of the following is an equation for the line tangent to the graph of f at the point where $x = 1$?
- (A) $y = 21x + 2$
- (B) $y = 21x - 19$ ✓
- (C) $y = 11x - 9$
- (D) $y = 10x + 2$
- (E) $y = 10x - 8$
25. An equation of the line tangent to the graph of $f(x) = x(1 - 2x)^3$ at the point $(1, -1)$ is
- (A) $y = -7x + 6$ ✓
- (B) $y = -6x + 5$
- (C) $y = -2x$
- (D) $y = 2x - 3$
- (E) $y = 7x - 8$
26. A particle moves along the x -axis so that at any time $t > 0$ its velocity is given by $v(t) = t \ln t - t$. At time $t = 1$, the position of the particle is $x(1) = 6$.
Write an expression for the acceleration of the particle.

Part A

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

✓

0

1

The student earns one of the following points:

1 point is earned for the correct $v'(t)$

$$a(t) = v'(t) = \ln t + t \cdot \frac{1}{t} - 1 = \ln t$$

2-8-8

27. A particle moves on the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = 2te^{-t}$.

Find the acceleration of the particle at $t=0$

Part A

1 point is earned for correctly finding $x'(t)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$

$$a(0) = x''(0) = -2 - 2 = -4$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for correctly finding $x'(t)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$

1 point is earned for correctly computing $x''(0)$

$$a(0) = x''(0) = -2 - 2 = -4$$

2-8-8

28. Particle X moves along the positive x -axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$.

A second particle, Y , moves along the positive y -axis so that its position at time t is given by $y(t) = 7t + 3$. At any time $t, t \geq 0$, the origin and the positions of the particles X and Y are the vertices of a triangle in the first quadrant. Find the rate of change of the area of the triangle at time $t = 1$. Show the work that leads to your answer.

Part C

The response can earn up to 4 points:

1 point: For the area function

$$A(t) = \frac{1}{2}x(t)y(t) = \frac{1}{2}(5t^3 - 9t^2 + 7)(7t + 3)$$

2 point: For the derivative

$$A'(t) = \frac{1}{2}[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 7)(7)]$$

1 point: For the correct answer

$$A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$$



0	1	2	3	4
---	---	---	---	---

The response earns all four of the following points:

1 point: For the area function

$$A(t) = \frac{1}{2}x(t)y(t) = \frac{1}{2}(5t^3 - 9t^2 + 7)(7t + 3)$$

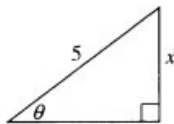
2 point: For the derivative

$$A'(t) = \frac{1}{2}[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 7)(7)]$$

1 point: For the correct answer

$$A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$$

29.



In the triangle shown above, if θ increases at a constant rate of 3 radians per minute, at what rate is x increasing in units per minute when x equals 3 units?

- (A) 3
 (B) $\frac{15}{4}$
 (C) 4
 (D) 9
 (E) 12



2-8-8

30. A particle moves along the x -axis so that at any time $t \geq 0$, its velocity is given by $v(t) = \sin(2t)$. If the position of the particle at time $t = \frac{\pi}{2}$ is $x = 4$, what is the particle's position at time $t = 0$?

(A) $-\frac{1}{2}$
 (B) 2
 (C) 3
 (D) 5
 (E) 8



31. How many critical points does the function $f(x) = (x + 2)^5(x - 3)^4$ have?

(A) One
 (B) Two
 (C) Three
 (D) Five
 (E) Nine



- 32.

X	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	-2	-3	4

The table above gives values of the differentiable functions f and g and their derivatives at $x = 1$. If $h(x) = (2f(x) + 3)(1 + g(x))$, then $h'(1) =$

(A) -28
 (B) -16
 (C) 40
 (D) 44
 (E) 47



33. If $y = x \sin x$, then $\frac{dy}{dx} =$

(A) $\sin x + \cos x$
 (B) $\sin x + x \cos x$
 (C) $\sin x - x \cos x$
 (D) $x(\sin x + \cos x)$
 (E) $x(\sin x - \cos x)$



34. Let f and g be differentiable functions with the following properties

(i) $g(x) > 0$ for all x

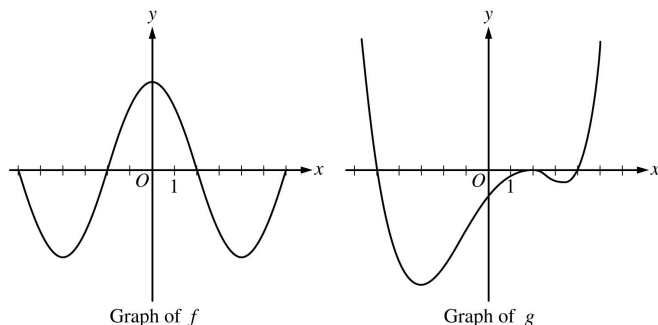
(ii) $f(0) = 1$ If $h(x) = f(x)g(x)$ and $h'(x) = f(x)g'(x)$, then $f(x) =$

(A) $f'(x)$
 (B) $g(x)$
 (C) e^x
 (D) 0
 (E) 1



2-8-8

35.



The graphs of two differentiable functions f and g are shown above. Given $p(x) = f(x)g(x)$ which of the following statements about $p'(-2)$ is true?

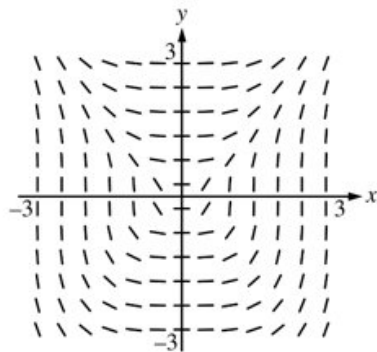
- (A) $p'(-2) < 0$ ✓
- (B) $p'(-2) = 0$
- (C) $p'(-2) > 0$
- (D) $p'(-2)$ is undefined.
- (E) There is not enough information given to conclude anything about $p'(-2)$.

36.

If $y = x^2 \sin 2x$, then $\frac{dy}{dx} =$

- (A) $2x \cos 2x$
- (B) $4x \cos 2x$
- (C) $2x (\sin 2x + \cos 2x)$
- (D) $2x (\sin 2x - x \cos 2x)$
- (E) $2x (\sin 2x + x \cos 2x)$ ✓

37.



Shown above is a slope field for which of the following differential equations?

- (A) $\frac{dy}{dx} = \frac{x}{y}$
- (B) $\frac{dy}{dx} = \frac{x^2}{y^2}$
- (C) $\frac{dy}{dx} = \frac{x^3}{y}$
- (D) $\frac{dy}{dx} = \frac{x^2}{y}$
- (E) $\frac{dy}{dx} = \frac{x^3}{y^2}$ ✓

2-8-8

38. If $f(x) = e^{2x}(x^3 + 1)$, then $f'(2) =$

- (A) $6e^4$
- (B) $21e^4$
- (C) $24e^4$
- (D) $30e^4$

**Answer D**

This option is correct. Use a combination of the product rule and the chain rule to compute the derivative, then evaluate that derivative at $x = 2$.

$$f'(x) = 2e^{2x}(x^3 + 1) + e^{2x}(3x^2)$$

$$f'(2) = 18e^4 + 12e^4 = 30e^4$$

2-8-8

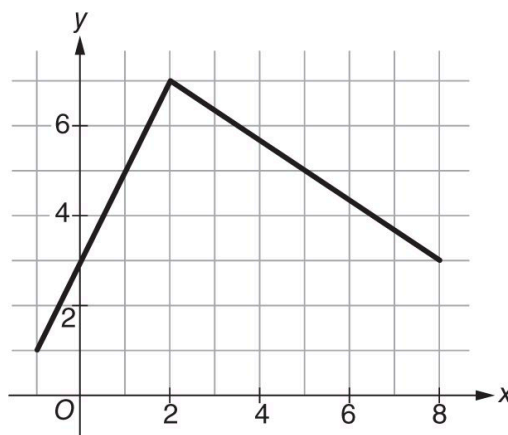
39.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be the continuous function defined on $[-1, 8]$ whose graph, consisting of two line segments, is shown above. Let g and h be the functions defined by $g(x) = \sqrt{x^2 - x + 3}$ and $h(x) = 5e^x - 9\sin x$.

- The function k is defined by $k(x) = f(x)g(x)$. Find $k'(0)$.
- The function m is defined by $m(x) = \frac{f(x)}{2g(x)}$. Find $m'(5)$.
- Find the value of x for $-1 < x < 2$ such that $f'(x) = h'(x)$.

Part A

The third point is earned for a correct product rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified. The third point can also be earned if a student imports an incorrect value for either $f'(0)$ or $g'(0)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $f'(0)$
- ☐ $g'(0)$
- ☐ $k'(0)$

2-8-8

Solution:

$$k'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$$f'(0) = 2 \text{ from the graph.}$$

$$g'(0) = -0.288675$$

$$\begin{aligned} k'(0) &= f'(0) \cdot g(0) + f(0) \cdot g'(0) \\ &= 2 \cdot \sqrt{3} + 3 \cdot (-0.288675) = 2.598 \end{aligned}$$

Part B

The third point is earned for a correct quotient rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $f'(5)$
- ☐ $g'(5)$
- ☐ $m'(5)$

Solution:

$$m'(x) = \frac{f'(x) \cdot 2g(x) - f(x) \cdot 2g'(x)}{(2g(x))^2}$$

$$f'(5) = -\frac{2}{3} \text{ from the graph.}$$

$$g'(5) = 0.938315$$

$$\begin{aligned} m'(5) &= \frac{f'(5) \cdot 2g(5) - f(5) \cdot 2g'(5)}{(2g(5))^2} \\ &= \frac{\left(-\frac{2}{3}\right) \cdot 2\sqrt{23} - 5 \cdot 2 \cdot 0.938315}{\left(2\sqrt{23}\right)^2} = -0.171 \end{aligned}$$

Part C

If $h'(x)$ has a maximum of one sign error with either term, the response is eligible for the second point based on a correct evaluation with the presented $h'(x)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

2-8-8

- ☐ $h'(x)$
- ☐ answer

Solution:

$$f'(x) = 2 \text{ for } -1 < x < 2$$

$$h'(x) = 5e^x - 9 \cos x$$

$$h'(x) = 2 \Rightarrow x = 0.622$$

2-8-8

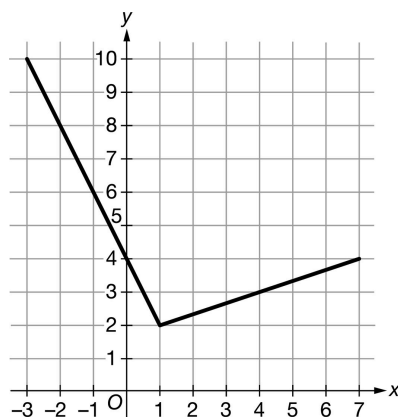
40.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be the continuous function defined on $[-3, 7]$ whose graph, consisting of two line segments, is shown above. Let g and h be the functions defined by $g(x) = (x^2 + 5x)^{\frac{1}{3}}$ and $h(x) = 7 \cos x + x^3$.

- The function N is defined by $N(x) = f(x)g(x)$. Find $N'(-1)$. Show the work that leads to your answer.
- The function P is defined by $P(x) = \frac{g(x)}{5f(x)}$. Find $P'(4)$. Show the work that leads to your answer.
- Find the value of x for $-3 < x < 1$ such that $f'(x) = h'(x)$.

Part A

The third point is earned for a correct product rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified. The third point can also be earned if a student imports an incorrect value for either $f'(-1)$ or $g'(-1)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 
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The student response accurately includes all three of the criteria below.

- ☐ $f'(-1)$
- ☐ $g'(-1)$
- ☐ $N'(-1)$

2-8-8

Solution:

$$N'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$f'(-1) = -2$ determined from the slope of the line segment on the graph.

$$g'(-1) = 0.396850$$

$$\begin{aligned} N'(-1) &= f'(-1) \cdot g(-1) + f(-1) \cdot g'(-1) \\ &= (-2) \cdot (-4)^{\frac{1}{3}} + 6 \cdot 0.396850 = 5.556 \text{ (or } 5.555) \end{aligned}$$

Part B

The third point is earned for a correct quotient rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

☐ $f'(4)$

☐ $g'(4)$

☐ $P'(4)$

Solution:

$$P'(x) = \frac{g'(x) \cdot 5f(x) - g(x) \cdot 5f'(x)}{(5f(x))^2}$$

$f'(4) = \frac{1}{3}$ determined from the slope of the line segment on the graph.

$$g'(4) = 0.397454$$

$$\begin{aligned} P'(4) &= \frac{g'(4) \cdot 5f(4) - g(4) \cdot 5f'(4)}{(5f(4))^2} \\ &= \frac{0.397454 \cdot 5 \cdot 3 - 36^{\frac{1}{3}} \cdot 5 \cdot \frac{1}{3}}{(5 \cdot 3)^2} = 0.002 \end{aligned}$$

Part C

If $h'(x)$ has a maximum of one sign error with either term, the response is eligible for the second point based on a correct evaluation with the presented $h'(x)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2
---	---	---

The student response accurately includes both of the criteria below.

2-8-8

- ☐ $h'(x)$
- ☐ answer

Solution:

$f'(x) = -2$ determined from the slope of the line segment on the graph for $-3 < x < 1$

$$h'(x) = -7 \sin x + 3x^2$$

$$h'(x) = -2 \Rightarrow x = 0.343 \text{ (or } 0.342)$$

2-8-8

41.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	$f(x)$	$f'(x)$
3	-1	8
5	4	3
7	6	-2

The table above gives values of the differentiable function f and its derivative f' at selected values of x . Let h be the function defined by $h(x) = x^2 e^{\cos(3x)}$.

- Using values of f from the table, estimate $f'(4)$. Show the work that leads to your answer.
- Let M be the function defined by $M(x) = f(x) + h(x)$. Find $M'(3)$.
- Let N be the function defined by $N(x) = f(x)h(x)$. Find $N'(5)$.
- Let P be the function defined by $P(x) = \frac{h(x)}{f(x)}$. Find $P'(7)$.

Part A

The point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified. If simplified, there are no errors. A response that presents an incorrect expression, makes an error in substituting values from the table, or makes an error in simplification does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct estimate of $f'(4)$

Solution:

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = \frac{4 - (-1)}{5 - 3} = \frac{5}{2}$$

Part B

The first point is earned for correctly presenting the rule for derivative of a sum, either by writing $M'(x) = f'(x) + h'(x)$ or $M'(3) = f'(3) + h'(3)$ or $M'(3) = 8 + (-2.061490)$ OR for the correct answer accompanied by $f'(3) = 8$ and $h'(3) = -2.061490$.

2-8-8

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $8 + (-2.061490)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $M'(x)$

☐ answer

Solution:

$$M'(x) = f'(x) + h'(x)$$

$$f'(3) = 8 \text{ from the table.}$$

$$h'(3) = -2.061490 \text{ from the calculator.}$$

$$M'(3) = f'(3) + h'(3) = 5.939 \text{ (or } 5.938)$$

Part C

The first point is earned for correctly presenting the product rule for derivatives, either by writing $N'(x) = f'(x)h(x) + f(x)h'(x)$ or $N'(5) = f'(5)h(5) + f(5)h'(5)$ or $N'(5) = 3(11.695310) + 4(-18.137830)$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $3(11.695310) + 4(-18.137830)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $N'(x)$

☐ answer

Solution:

$$N'(x) = f'(x)h(x) + f(x)h'(x)$$

$$f(5) = 4 \text{ and } f'(5) = 3 \text{ from the table.}$$

$$h(5) = 11.695310 \text{ and } h'(5) = -18.137830 \text{ from the calculator.}$$

$$N'(5) = f'(5)h(5) + f(5)h'(5) = -37.465$$

Part D

2-8-8

The first point is earned for correctly presenting the quotient rule for derivatives, either by writing $P'(x) = \frac{h'(x)f(x) - h(x)f'(x)}{(f(x))^2}$ or

$$P'(7) = \frac{h'(7)f(7) - h(7)f'(7)}{(f(7))^2} \text{ or } P'(7) = \frac{(-63.023773) \cdot 6 - 28.334809 \cdot (-2)}{6^2}.$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term in the numerator and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $\frac{(-63.023773) \cdot 6 - 28.334809 \cdot (-2)}{6^2}$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $P'(x)$

☐ answer

Solution:

$$P'(x) = \frac{h'(x)f(x) - h(x)f'(x)}{(f(x))^2}$$

$f(7) = 6$ and $f'(7) = -2$ from the table.

$h(7) = 28.334809$ and $h'(7) = -63.023773$ from the calculator.

$$P'(7) = \frac{h'(7)f(7) - h(7)f'(7)}{(f(7))^2} = -8.930 \text{ (or } -8.929)$$

2-8-8

42. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	-2	-1	1
$f(x)$	$-\frac{5}{2}$	-3	-2	$\frac{2}{3}$
$f'(x)$	-1	$\frac{1}{3}$	$\frac{6}{5}$	$\frac{4}{3}$

The table above gives values of the differentiable function f and its derivative for selected values of x .

- (a) Let g be the function defined by $g(x) = \frac{f(x^2)}{e^x}$. Find $g'(-1)$.
- (b) Let h be the function defined by $h(x) = f(f(-2x))$. Find $h'(1)$.
- (c) Let k be the function defined by $k(x) = f(x) \cdot \arcsin\left(\frac{x}{2}\right)$. Find $k'(-1)$.

Part A

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student responses accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ answer

Solution:

$$g'(x) = \frac{f'(x^2) \cdot 2x \cdot e^x - f(x^2) \cdot e^x}{(e^x)^2}$$

$$g'(-1) = \frac{f'(1) \cdot 2(-1) \cdot e^{-1} - f(1) \cdot e^{-1}}{(e^{-1})^2}$$

$$= \frac{\frac{4}{3}(-2)e^{-1} - \frac{2}{3}e^{-1}}{(e^{-1})^2} = -\frac{10e}{3}$$

Part B

2-8-8

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $h'(x)$
- ☐ answer

Solution:

$$h'(x) = f'(f(-2x)) \cdot f'(-2x) \cdot (-2)$$

$$\begin{aligned} h'(1) &= f'(f(-2)) \cdot f'(-2) \cdot (-2) \\ &= f'(-3) \cdot \frac{1}{3} \cdot (-2) = (-1) \cdot \frac{1}{3} \cdot (-2) = \frac{2}{3} \end{aligned}$$

Part C

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ answer

Solution:

$$k'(x) = f'(x) \cdot \arcsin\left(\frac{x}{2}\right) + f(x) \cdot \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}} \cdot \frac{1}{2}$$

$$\begin{aligned} k'(-1) &= f'(-1) \cdot \arcsin\left(\frac{-1}{2}\right) + f(-1) \cdot \frac{1}{\sqrt{1-\left(\frac{-1}{2}\right)^2}} \cdot \frac{1}{2} \\ &= \frac{6}{5} \cdot \left(-\frac{\pi}{6}\right) + (-2) \cdot \frac{2}{\sqrt{3}} \cdot \frac{1}{2} = -\frac{\pi}{5} - \frac{2}{\sqrt{3}} \end{aligned}$$

2-8-8

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	2	3	8
$f(x)$	-9	4	2	6
$f'(x)$	$-\frac{7}{2}$	$\frac{3}{2}$	$-\frac{2}{5}$	$\frac{1}{3}$

The table above gives values of the differentiable function f and its derivative for selected values of x .

- (a) Let g be the function defined by $g(x) = \frac{\ln x}{f(x^3)}$. Find $g'(2)$.
- (b) Let h be the function defined by $h(x) = f(f(-3x))$. Find $h'(-1)$.
- (c) Let k be the function defined by $k(x) = f(x) \cdot \arctan\left(\frac{x}{3}\right)$. Find $k'(-3)$.

Part A

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

				✓
0	1	2	3	

The student responses accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ answer

Solution:

$$g'(x) = \frac{f(x^3) \cdot \frac{1}{x} - f'(x^3) \cdot 3x^2 \cdot \ln x}{(f(x^3))^2}$$

$$g'(2) = \frac{f(8) \cdot \frac{1}{2} - f'(8) \cdot 12 \ln 2}{(f(8))^2}$$

$$= \frac{6 \cdot \frac{1}{2} - \frac{1}{3} \cdot 12 \ln 2}{(6)^2} = \frac{3 - 4 \ln 2}{36}$$

Part B

2-8-8

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response does not accurately include any of the criteria below.

- ☐ $h'(x)$
- ☐ answer

Solution:

$$h'(x) = f'(f(-3x)) \cdot f'(-3x) \cdot (-3)$$

$$h'(-1) = f'(f(3)) \cdot f'(3) \cdot (-3)$$

$$= f'(2) \cdot \left(-\frac{2}{5}\right) \cdot (-3) = \left(\frac{3}{2}\right) \cdot \left(-\frac{2}{5}\right) \cdot (-3) = \frac{9}{5}$$

Part C

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ answer

Solution:

$$k'(x) = f'(x) \cdot \arctan\left(\frac{x}{3}\right) + f(x) \cdot \frac{1}{1+\left(\frac{x}{3}\right)^2} \cdot \frac{1}{3}$$

$$k'(-3) = f'(-3) \cdot \arctan(-1) + f(-3) \cdot \frac{1}{1+\left(\frac{-3}{3}\right)^2} \cdot \frac{1}{3}$$

$$= -\frac{7}{2} \cdot \left(-\frac{\pi}{4}\right) + (-9) \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{7\pi}{8} - \frac{3}{2}$$

2-8-8

44.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	$g(x)$	$g'(x)$
2	-2	4
4	5	1
6	2	-3

The table above gives values of the differentiable function g and its derivative g' at selected values of x . Let f be the function defined by $f(x) = x \sin(\sqrt{3x^2 + 5})$.

- Using values of g from the table, estimate $g'(5)$. Show the work that leads to your answer.
- Let R be the function defined by $R(x) = f(x) - g(x)$. Find $R'(2)$.
- Let S be the function defined by $S(x) = f(x)g(x)$. Find $S'(4)$.
- Let T be the function defined by $T(x) = \frac{f(x)}{g(x)}$. Find $T'(6)$.

Part A

The point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified. If simplified, there are no errors. A response that presents an incorrect expression, makes an error in substituting values from the table, or makes an error in simplification does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes a correct estimate of $g'(5)$

Solution:

$$g'(5) \approx \frac{g(6) - g(4)}{6 - 4} = \frac{2 - 5}{6 - 4} = -\frac{3}{2}$$

Part B

The first point is earned for correctly presenting the rule for derivative of a difference, either by writing $R'(x) = f'(x) - g'(x)$ or $R'(2) = f'(2) - g'(2)$ or $R'(2) = -2.448854 - 4$ OR for the correct answer accompanied by $f'(2) = -2.448854$ and $g'(2) = 4$.

2-8-8

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $-2.448854 - 4$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

☐ $R'(x)$

☐ answer

Solution:

$$R'(x) = f'(x) - g'(x)$$

$$f'(2) = -2.448854 \text{ from the calculator.}$$

$$g'(2) = 4 \text{ from the table.}$$

$$R'(2) = f'(2) - g'(2) = -6.449$$

Part C

The first point is earned for correctly presenting the product rule for derivatives, either by writing $S'(x) = f'(x)g(x) + f(x)g'(x)$ or $S'(4) = f'(4)g(4) + f(4)g'(4)$ or

$$S'(4) = (4.419230) \cdot (5) + (3.359221) \cdot (1).$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as

$$(4.419230) \cdot (5) + (3.359221) \cdot (1) \text{ earns the second point.}$$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

☐ $S'(x)$

☐ answer

Solution:

$$S'(x) = f'(x)g(x) + f(x)g'(x)$$

$$f(4) = 3.359221 \text{ and } f'(4) = 4.419230 \text{ from the calculator.}$$

$$g(4) = 5 \text{ and } g'(4) = 1 \text{ from the table.}$$

$$S'(4) = f'(4)g(4) + f(4)g'(4) = 25.455$$

2-8-8

Part D

The first point is earned for correctly presenting the quotient rule for derivatives, either by writing $T'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$ or

$$T'(6) = \frac{f'(6)g(6) - f(6)g'(6)}{(g(6))^2} \text{ or}$$

$$T'(6) = \frac{(-4.564565) \cdot (2) - (-5.603824) \cdot (-3)}{(2)^2}.$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term in the numerator and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as

$$\frac{(-4.564565) \cdot (2) - (-5.603824) \cdot (-3)}{(2)^2} \text{ earns the second point.}$$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓		
0	1	2

The student response accurately includes both of the criteria below.

☐ $T'(x)$

☐ answer

Solution:

$$T'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$f(6) = -5.603824 \text{ and}$$

$$f'(6) = -4.564565 \text{ from the calculator.}$$

$$g(6) = 2 \text{ and } g'(6) = -3 \text{ from the table.}$$

$$T'(6) = \frac{f'(6)g(6) - f(6)g'(6)}{(g(6))^2} = -6.485$$

45. If $y = \sin x \cos x$, then at $x = \frac{\pi}{3}$, $\frac{dy}{dx} =$

(A) $-\frac{1}{2}$

(B) $-\frac{1}{4}$

(C) $\frac{1}{4}$

(D) $\frac{1}{2}$

(E) 1



2-8-8

46. The function f is differentiable for all real numbers. The point $(3, \frac{1}{4})$ is on the graph of $y = f(x)$, and the slope at each point (x, y) on the graph is given by $\frac{dy}{dx} = y^2(6 - 2x)$.

(a) Find $\frac{d^2y}{dx^2}$ and evaluate it at the point $(3, \frac{1}{4})$.

(b) Find $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = y^2(6 - 2x)$ with the initial condition $f(3) = 1/4$.

Part A

2 points are earned for $\frac{d^2y}{dx^2}$

<-2> product rule or chain rule error

1 point is earned for the value at $(3, \frac{1}{4})$

$$\begin{aligned}\frac{d^2y}{dx^2} &= 2y \frac{dy}{dx} (6 - 2x) - 2y^2 \\ &= 2y^3(6 - 2x)^2 - 2y^2 \\ \frac{d^2y}{dx^2} \Big|_{(3, \frac{1}{4})} &= 0 - 2\left(\frac{1}{4}\right)^2 = -\frac{1}{8}\end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

<-2> product rule or chain rule error

1 point is earned for the value at $(3, \frac{1}{4})$

$$\begin{aligned}\frac{d^2y}{dx^2} &= 2y \frac{dy}{dx} (6 - 2x) - 2y^2 \\ &= 2y^3(6 - 2x)^2 - 2y^2 \\ \frac{d^2y}{dx^2} \Big|_{(3, \frac{1}{4})} &= 0 - 2\left(\frac{1}{4}\right)^2 = -\frac{1}{8}\end{aligned}$$

Part B

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $f(3) = 1/4$

1 point is earned for solves for y

2-8-8

$$\frac{1}{y^2} dy = (6 - 2x) dx$$

$$-\frac{1}{y} = 6x - x^2 + C$$

$$-4 = 18 - 9 + C = 9 + C$$

$$C = -13$$

$$y = \frac{1}{x^2 - 6x + 13}$$



0	1	2	3	4	5	6
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The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivative of dy term

1 point is earned for the antiderivative of dx term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $f(3) = 1/4$

1 point is earned for solves for y

$$\frac{1}{y^2} dy = (6 - 2x) dx$$

$$-\frac{1}{y} = 6x - x^2 + C$$

$$-4 = 18 - 9 + C = 9 + C$$

$$C = -13$$

$$y = \frac{1}{x^2 - 6x + 13}$$

2-8-8

47. The function f is differentiable for all real numbers. The point $(3, \frac{1}{4})$ is on the graph of $y = f(x)$, and the slope at each point (x, y) on the graph is given by $\frac{dy}{dx} = y^2(6 - 2x)$.

Find $\frac{d^2y}{dx^2}$ and evaluate it at the point $(3, \frac{1}{4})$.

Part A

2 points are earned for $\frac{d^2y}{dx^2}$

< -2 > product rule or chain rule error

1 point is earned for the value at $(3, \frac{1}{4})$

$$\frac{d^2y}{dx^2} = 2y \frac{dy}{dx} (6 - 2x) - 2y^2$$

$$= 2y^3(6 - 2x)^2 - 2y^2$$

$$\frac{d^2y}{dx^2} \Big|_{(3, \frac{1}{4})} = 0 - 2\left(\frac{1}{4}\right)^2 = -\frac{1}{8}$$



0	1	2	3
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The student response earns three of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

< -2 > product rule or chain rule error

1 point is earned for the value at $(3, \frac{1}{4})$

$$\frac{d^2y}{dx^2} = 2y \frac{dy}{dx} (6 - 2x) - 2y^2$$

$$= 2y^3(6 - 2x)^2 - 2y^2$$

$$\frac{d^2y}{dx^2} \Big|_{(3, \frac{1}{4})} = 0 - 2\left(\frac{1}{4}\right)^2 = -\frac{1}{8}$$

2-8-8

48. The derivative of a function f is given by $f'(x) = (x - 3)e^x$ for $x > 0$, and $f(1) = 7$.
On what intervals, if any, is the graph of f both decreasing and concave up? Explain your reasoning.

Part B

The response can earn up to 3 points:

2 point: For $f''(x)$

1 point: For correct answer with reason

$$f''(x) = e^x + (x - 3)e^x = (x - 2)e^x$$

$$f''(x) > 0 \text{ for } x > 2$$

$$f'(x) < 0 \text{ for } 0 < x < 3$$

Therefore, the graph of f is both decreasing and concave up on the interval $2 < x < 3$.



0	1	2	3
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The response earns all three of the following points:

2 point: For $f''(x)$

1 point: For correct answer with reason

$$f''(x) = e^x + (x - 3)e^x = (x - 2)e^x$$

$$f''(x) > 0 \text{ for } x > 2$$

$$f'(x) < 0 \text{ for } 0 < x < 3$$

Therefore, the graph of f is both decreasing and concave up on the interval $2 < x < 3$.

2-8-8

49. The twice-differentiable function f is defined for all real numbers and satisfies the following conditions: $f(0) = 2$, $f'(0) = -4$, and $f''(0) = 3$.

The function h is given by $h(x) = \cos(kx)f(x)$ for all real numbers, where k is a constant. Find $h'(x)$ and write an equation for the line tangent to the graph of h at $x = 0$.

Part B

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h'(0) = \cos(0) f'(0) = -4$$

The equation of the tangent line is $y = -4x + 2$.



0	1	2	3	4	5
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The student response earns all of the following points:

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h'(0) = \cos(0) f'(0) = -4$$

The equation of the tangent line is $y = -4x + 2$.

50. If $f(x) = (x - 1)^2 \sin x$, then $f'(0) =$

(A) -2

(B) -1

(C) 0

(D) 1

(E) 2



2-8-8

51. If $y = x^2 e^x$, then $\frac{dy}{dx} =$

(A) $2xe^x$

(B) $x(x + 2e^x)$

(C) $xe^x(x + 2)$ ✓

(D) $2x + e^x$

(E) $2x + e$

52. The slope of the line tangent to the graph of $y = xe^x$ at $x = \ln 2$ is

(A) $2 \ln 2$

(B) $2 \ln 2 + 2$ ✓

(C) $e^2(\ln 2) + e^2$

(D) $2 + \frac{2 \ln 2}{e}$

53. What are all values of x for which the function f defined by $f(x) = (x^2 - 3)e^{-x}$ is increasing?(A) There are no such values of x .

(B) $x < -1$ and $x > 3$

(C) $-3 < x < 1$

(D) $-1 < x < 3$ ✓

(E) All values of x 54. If $y = x^2(e^x - 1)$, then $\frac{dy}{dx} =$

(A) $2xe^x$

(B) $2xe^x - 2x$

(C) $x^2e^x + 2xe^x - 2x$ ✓

(D) $x^2e^x + 2xe^x - x^2 - 2x$

Answer C

55. If $y = x\sqrt{2x+5}$, then $y' =$

(A) $\frac{3x+5}{\sqrt{2x+5}}$ ✓

(B) $\frac{1}{\sqrt{2x+5}}$

(C) $\frac{1}{2\sqrt{2x+5}}$

(D) $\frac{5x+10}{2\sqrt{2x+5}}$

2-8-8

56. If $y = 5x\sqrt{x^2 + 1}$, then $\frac{dy}{dx}$ at $x = 3$ is

(A) $\frac{5}{2\sqrt{10}}$

(B) $\frac{15}{\sqrt{10}}$

(C) $\frac{15}{2\sqrt{10}} + 5\sqrt{10}$

(D) $\frac{45}{\sqrt{10}} + 5\sqrt{10}$ ✓

(E) $\frac{45}{\sqrt{10}} + 15\sqrt{10}$

2-8-8

57. Consider the logistic differential equation $\frac{dy}{dt} = \frac{y}{8}(6 - y)$. Let $y = f(t)$ be the particular solution to the differential equation with $f(0) = 8$.

Write the second-degree Taylor polynomial for f about $t = 0$, and use it to approximate $f(1)$.

Part C

The response can earn up to 4 points:

2 points: $\frac{d^2y}{dt^2}$

1 point: second-degree Taylor polynomial

1 point: approximation of $f(1)$

$$\frac{d^2y}{dt^2} = \frac{1}{8} \frac{dy}{dt} (6 - y) + \frac{y}{8} \left(-\frac{dy}{dt} \right) f(0) = 8; \quad f'(0) = \frac{dy}{dt} \bigg|_{t=0} = \frac{8}{8}(6 - 8) = -2; \quad \text{and} \quad f''(0) = \frac{d^2y}{dt^2} \bigg|_{t=0} = \frac{1}{8}(-2)(-2) + \frac{8}{8}(2) = \frac{5}{2}$$

The second-degree Taylor polynomial for f about

$$t = 0 \text{ is } P_2(t) = 8 - 2t + \frac{5}{4}t^2. \quad f(1) \approx P_2(1) = \frac{29}{4}$$



0	1	2	3	4
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The response can earn up to 4 points:

2 points: $\frac{d^2y}{dt^2}$

1 point: second-degree Taylor polynomial

1 point: approximation of $f(1)$

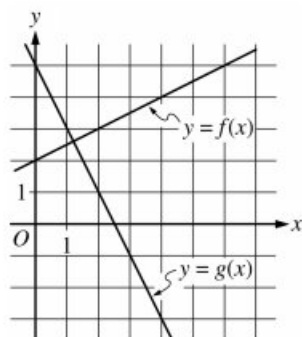
$$\frac{d^2y}{dt^2} = \frac{1}{8} \frac{dy}{dt} (6 - y) + \frac{y}{8} \left(-\frac{dy}{dt} \right) f(0) = 8; \quad f'(0) = \frac{dy}{dt} \bigg|_{t=0} = \frac{8}{8}(6 - 8) = -2; \quad \text{and} \quad f''(0) = \frac{d^2y}{dt^2} \bigg|_{t=0} = \frac{1}{8}(-2)(-2) + \frac{8}{8}(2) = \frac{5}{2}$$

The second-degree Taylor polynomial for f about

$$t = 0 \text{ is } P_2(t) = 8 - 2t + \frac{5}{4}t^2. \quad f(1) \approx P_2(1) = \frac{29}{4}$$

2-8-8

58.



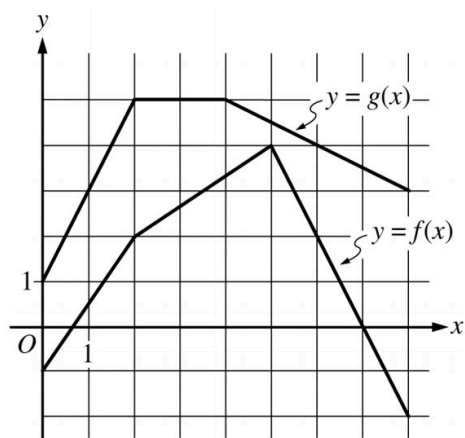
The figure above shows the graphs of the functions f and g .

If $h(x) = f(x)g(x)$, then $h'(2) =$

- (A) $\frac{13}{2}$
 (B) $\frac{1}{2}$
 (C) -1
 (D) $-\frac{11}{2}$



59.



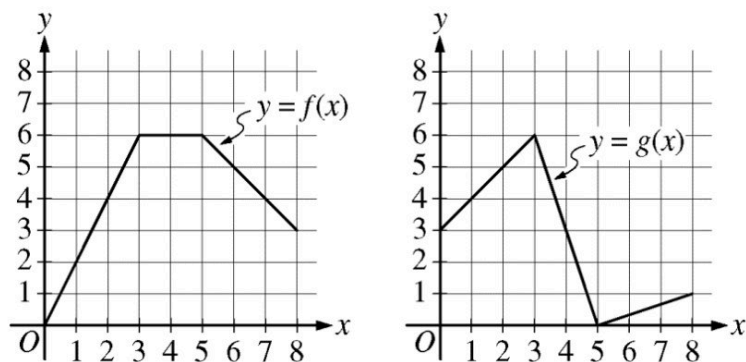
The graphs of f and g are shown above. If $h(x) = f(x)g(x)$, then $h'(6) =$

- (A) -9
 (B) -7
 (C) 1
 (D) 7
 (E) 9



2-8-8

60.

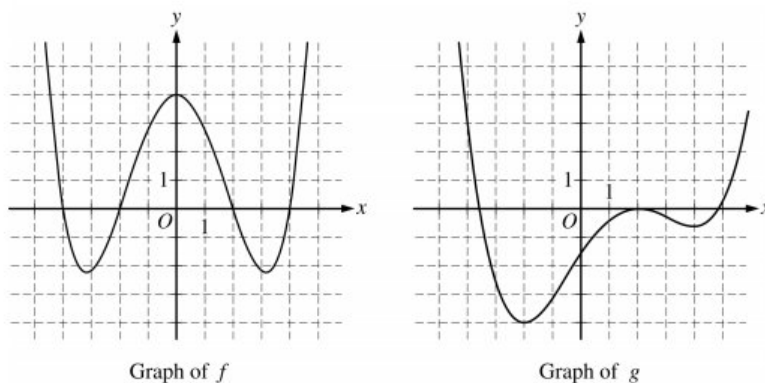


The graphs of the piecewise linear functions f and g are shown above. If the function h is defined by $h(x) = f(x)g(x)$, then $h'(2)$ is

- (A) 2
- (B) 13
- (C) 14
- (D) 20
- (E) nonexistent



61.



The graphs of the differentiable functions f and g are shown above. If the function p is defined by $p(x) = f(x)g(x)$, which of the following must be true about p' , the derivative of p ?

- (A) $p'(-2) < 0$
- (B) $p'(-2) = 0$
- (C) $p'(-2) > 0$
- (D) $p'(0) < 0$
- (E) $p'(0) = 0$



2-8-8

62. The twice-differentiable function f is defined for all real numbers and satisfies the following conditions:

$$f(0) = 2, f'(0) = -4, \text{ and } f''(0) = 3.$$

- (a) The function g is given by $g(x) = e^{ax} + f(x)$ for all real numbers, where a is a constant. Find $g'(0)$ and $g''(0)$ in terms of a . Show the work that leads to your answer.
- (b) The function h is given by $h(x) = \cos(kx)f(x)$ for all real numbers, where k is a constant. Find $h'(x)$ and write an equation for the line tangent to the graph of h at $x = 0$.

Part A

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = ae^{ax} + f'(x)$$

$$g'(0) = a - 4$$

$$g''(x) = a^2e^{ax} + f''(x)$$

$$g''(0) = a^2 + 3$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: $g'(x)$

1 point is earned for: $g'(0)$

1 point is earned for: $g''(x)$

1 point is earned for: $g''(0)$

$$g'(x) = ae^{ax} + f'(x)$$

$$g'(0) = a - 4$$

$$g''(x) = a^2e^{ax} + f''(x)$$

$$g''(0) = a^2 + 3$$

Part B

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

2-8-8

$$h(0) = \cos(0)f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.



0	1	2	3	4	5
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The student response earns all of the following points:

2 points are earned for: $h'(x)$

1 point is earned for: $h'(0)$

1 point is earned for: $h(0)$

1 point is earned for the equation of tangent line

$$h'(x) = f'(x) \cos(kx) - k \sin(kx) f(x)$$

$$h'(0) = f'(0) \cos(0) - k \sin(0) f(0) = f'(0) = -4$$

$$h(0) = \cos(0)f(0) = 2$$

The equation of the tangent line is $y = -4x + 2$.

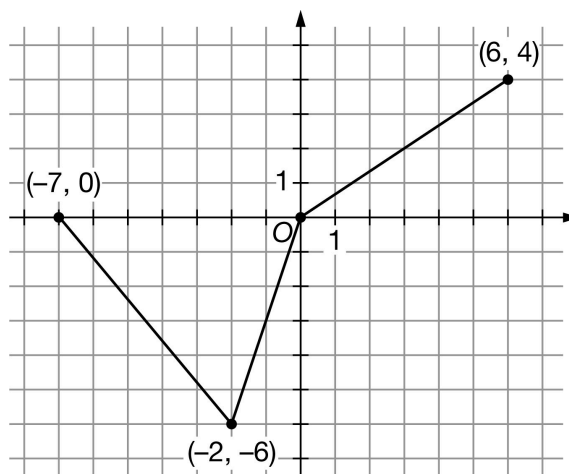
2-8-8

63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. It is known that $h(-7) = 2$.

- Find the x -coordinate of each critical point of h on the open interval $-7 < x < 6$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-7 \leq x \leq 6$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(-2) = -13$. Find the value of $g'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-7 < x < 6$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-7 < x < 6$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-7 < x < 6$; this work does not affect whether the point is earned.

2-8-8



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 0$
- the slope of h changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$
- Any incorrect argument

Note: If additional x -coordinates in $-7 < x < 6$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-7 < x < 6$ are identified in part (a), do not read reasons for these responses. Only read for values in the interval.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

2-8-8

First point: Considers both endpoints

Earns the first point by identifying $x = -7$ and $x = 6$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-7) = 2$ in a calculation for $h(0)$ and/or $h(6)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 2.

The ordered pair $(-7, 2)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-7	2
0	-19
6	-7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-7 < x < 6$ is a relative minimum, so the absolute maximum value of h on $-7 \leq x \leq 6$ must occur at an endpoint.

$$h(-7) = 2$$

$$\begin{aligned} h(6) &= h(-7) + \int_{-7}^6 h'(t) dt \\ &= 2 + \int_{-7}^0 h'(t) dt + \int_0^6 h'(t) dt \\ &= 2 - \frac{1}{2} \cdot 7 \cdot 6 + \frac{1}{2} \cdot 6 \cdot 4 = -7 \end{aligned}$$

Therefore, the absolute maximum value of h on $-7 \leq x \leq 6$ is $h(-7) = 2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' changes from negative to positive at $x = -2$
- the slope of h' changes from negative to positive at $x = -2$

2-8-8

- h'' changes sign at $x = -2$
- a relative extrema exists on the graph of h' at $x = -2$
- a relative minimum exists on the graph of h' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of h'']
- h' changes from negative to positive
- the graph changes from negative to positive
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = -2$
- The acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is increasing for $x < 0$ and $x > 0$
- h' is increasing through $x = 0$
- h'' is always positive before and after $x = 0$
- the slope of h' is always positive before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of h'']
- h' is always positive
- h' does not change sign

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = 0$ is not the location a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow 3} (2h'(x) - 4) = 0$ and $\lim_{x \rightarrow 3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

· $\lim_{x \rightarrow 3} (2h'(x) - 4) = 0$ and $\lim_{x \rightarrow 3} (x^2 - 9) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} \rightarrow 0$

· $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow 3} \frac{2h'(x)}{2x}$

· $\frac{2h''(3)}{2(3)}$

· $\frac{2h'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $\frac{2}{9}$ or equivalent



0	1	2	3
---	---	---	---

2-8-8

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = 3$. $\Rightarrow \lim_{x \rightarrow 3} h'(x) = h'(3)$ and $\lim_{x \rightarrow 3} h''(x) = h''(3)$.

$$\lim_{x \rightarrow 3} 2h'(x) - 4 = 2 \cdot 2 - 4 = 0$$

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} &= \lim_{x \rightarrow 3} \frac{2h''(x)}{2x} \\ &= \frac{h''(3)}{3} = \frac{\frac{2}{3}}{3} = \frac{2}{9} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$

Product rule may appear as $g'(-2) = -2 \cdot h'(-2) + h(-2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $g'(-2) = -1$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(-2) = -2 \cdot h'(-2) + h(-2) = -2 \cdot (-6) + (-13) = -1$$

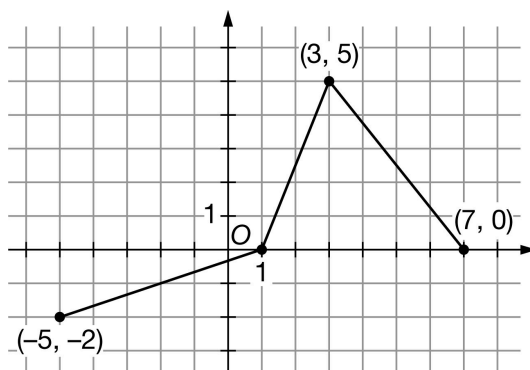
2-8-8

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-5 \leq x \leq 7$ consists of three line segments, as shown in the figure above. It is known that $h(-5) = 6$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 7$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 7$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 1$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(3) = 5$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 1$ only
- $x = 1$ and any additional x -coordinates not in $-5 < x < 7$

Does not earn the point:

- $x = 1$ and any additional x -coordinates in $-5 < x < 7$
- $x = 1$ with any incorrect supporting work

Do not read work related to additional values not in $-5 < x < 7$; this work does not affect whether the point is earned.

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 1$
- the slope of h changes from negative to positive at $x = 1$
- the graph changes from negative to positive at $x = 1$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 1$ and increasing to the right of $x = 1$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 1$

Note: If additional x -coordinates in $-5 < x < 7$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-5 < x < 7$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 1$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

2-8-8

Earns the first point by identifying $x = -5$ and $x = 7$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-5) = 6$ in calculation for $h(1)$ or $h(7)$

Third point: Answer with justification

The response must declare as an answer the absolute maximum value of 15.

The ordered pair $(7, 15)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-5	6
1	0
7	15



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-5 < x < 7$ is a relative minimum, so the absolute maximum value of h on $-5 \leq x \leq 7$ must occur at an endpoint.

$$h(-5) = 6$$

$$\begin{aligned} h(7) &= h(-5) + \int_{-5}^7 h'(t) \, dt \\ &= 6 + \int_{-5}^1 h'(t) \, dt + \int_1^7 h'(t) \, dt \\ &= 6 - \frac{1}{2} \cdot 6 \cdot 2 + \frac{1}{2} \cdot 6 \cdot 5 = 15 \end{aligned}$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 7$ is $h(7) = 15$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from increasing to decreasing
- h'' changes from positive to negative at $x = 3$
- the slope of h' changes from positive to negative at $x = 3$
- h'' changes sign at $x = 3$

2-8-8

- a relative extrema exists on the graph of h' at $x = 3$
- a relative maximum exists on the graph of h' at $x = 2$

Does not earn the point:

- the concavity changes at $x = 3$ [Must appeal to the change in sign of h'']
- h' changes from positive to negative
- the graph changes from positive to negative
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of h' at $x = 2$
- the acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from increasing to decreasing there .

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 1$ is not the location of a point of inflection because

- h' is increasing for $x < 1$ and $x > 1$
- h' is increasing through $x = 1$
- h'' is always positive before and after $x = 1$
- the slope of h' is always positive before and after $x = 1$
- h'' does not change sign at $x = 1$
- a relative extrema does not exist on the graph of h' at $x = 1$

Does not earn the point:

- the concavity does not change [Must appeal to the change in sign of h'']
- h' is always positive
- h' does not change sign



0	1
---	---

2-8-8

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 1$ is not the location of a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (h'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

· $\lim_{x \rightarrow -2} (h'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} \rightarrow \frac{0}{0}$

· $\lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow -2} \frac{h'(x)}{2x}$

· $\frac{h''(-2)}{2(-2)}$

· $\frac{h'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $-\frac{1}{12}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule

2-8-8

☐ Answer**Solution:**

h' and h'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$ and $\lim_{x \rightarrow -2} h''(x) = h''(-2)$.

$$\lim_{x \rightarrow -2} h'(x) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \lim_{x \rightarrow -2} \frac{h''(x)}{2x}$$

$$= \frac{h''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{3}}{-4} = -\frac{1}{12}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

1st point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$

Product rule may appear as $g'(3) = 3 \cdot h'(3) + h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the 1st point

Must be our answer, $g'(3) = 20$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(3) = 3 \cdot h'(3) + h(3) = 3 \cdot 5 + 5 = 20$$

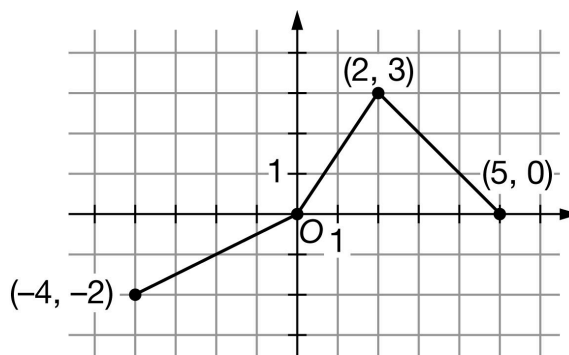
2-8-8

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-4 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-4) = 7$.

- Find the x -coordinate of each critical point of g on the open interval $-4 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute maximum value of g on the interval $-4 \leq x \leq 5$. Justify your answer.
- Is $x = 2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{g'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = x \cdot g(x)$. It is known that $g(2) = 6$. Find the value of $h'(2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-4 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-4 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-4 < x < 5$; this work does not affect whether the point is earned.

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $g'(x)$ changes from negative to positive at $x = 0$
- the slope of g changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- g is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-4 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is NOT eligible to earn the point.

Note: If additional x -coordinates not in $-4 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for g because $g'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

2-8-8

Earns the first point by identifying $x = -4$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-4) = 7$ in a calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 10.5.

The ordered pair $(5, 10.5)$ is not a declaration of the maximum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-4	7
0	3
6	10.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-4 < x < 5$ is a relative minimum, so the absolute maximum value of g on $-4 \leq x \leq 5$ must occur at an endpoint.

$$g(-4) = 7$$

$$\begin{aligned} g(5) &= g(-4) + \int_{-4}^5 g'(t) \, dt \\ &= 7 + \int_{-4}^0 g'(t) \, dt + \int_0^5 g'(t) \, dt \\ &= 7 - \frac{1}{2} \cdot 4 \cdot 2 + \frac{1}{2} \cdot 3 \cdot 5 = 10.5 \end{aligned}$$

Therefore, the absolute maximum value of g on $-4 \leq x \leq 5$ is $g(5) = 10.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 2$ is the location of a point of inflection because

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = 2$
- the slope of g' changes from positive to negative at $x = 2$
- g'' changes sign at $x = 2$

2-8-8

- a relative extrema exists on the graph of g' at $x = 2$
- a relative maximum exists on the graph of g' at $x = 2$

Does not earn the point:

- the concavity at $x = 2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ or $g''(x)$ does not exist (DNE)
- a relative minimum exists on the graph of g' at $x = 2$
- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is increasing for $x < 0$ and $x > 0$
- g' is increasing through $x = 0$
- g'' is always positive before and after $x = 0$
- the slope of g' is always positive before and after $x = 0$
- g'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always positive
- g' does not change sign



0	1
---	---

2-8-8

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of g because g' is increasing on both sides of $x = 0$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (g'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -2$

Earns the point:

· $\lim_{x \rightarrow -2} (g'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} \rightarrow \frac{0}{0}$

· $\lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow -2} \frac{g'(x)}{2x}$

· $\frac{g''(-2)}{2(-2)}$

· $\frac{g'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $-\frac{1}{8}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Limits of numerator and denominator equal 0

☐ Application of L'Hospital's Rule

2-8-8

☐ Answer**Solution:**

g' and g'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} g'(x) = g'(-2)$ and $\lim_{x \rightarrow -2} g''(x) = g''(-2)$.

$$\lim_{x \rightarrow -2} (g'(x) + 1) = g'(-2) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \lim_{x \rightarrow -2} \frac{g''(x)}{2x}$$

$$= \frac{g''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{2}}{-4} = -\frac{1}{8}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product rule

Applies product rule: $h'(x) = x \cdot g'(x) + g(x)$

Product rule may appear as $h'(2) = 2 \cdot g'(2) + g(2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(2) = 12$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$h'(x) = x \cdot g'(x) + g(x)$$

$$h'(2) = 2 \cdot g'(2) + g(2) = 2 \cdot 3 + 6 = 12$$

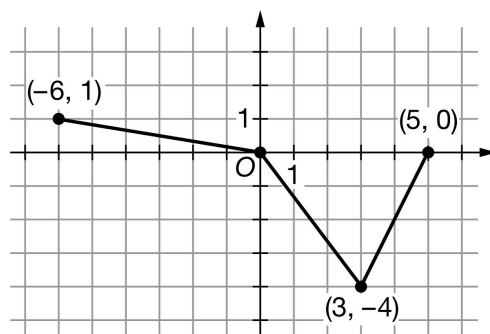
2-8-8

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $h(-6) = 5$.

- Find the x -coordinate of each critical point of h on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute minimum value of h on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -3} \frac{2h'(x)-1}{x^2-9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = 2x \cdot h(x)$. It is known that $h(3) = 2$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-6 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $h'(x)$ changes from positive to negative at $x = 0$
- the slope of h changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- h is increasing to the left of $x = 0$
- and decreasing to the right of $x = 0$
- (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for h because $h'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-8-8

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the 1st point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-6) = 5$ in a calculation for $h(0)$ or $h(5)$

Third point: Answer with justification

The response must declare as an answer the minimum value of -2 .

The ordered pair $(5, -2)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-6	5
0	8
5	-2



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of h on $-6 \leq x \leq 5$ must occur at an endpoint.

$$h(-6) = 5$$

$$\begin{aligned} h(5) &= h(-6) + \int_{-6}^5 h'(t) dt \\ &= 5 + \int_{-6}^0 h'(t) dt + \int_0^5 h'(t) dt \\ &= 5 + \frac{1}{2} \cdot 6 \cdot 1 - \frac{1}{2} \cdot 5 \cdot 4 = -2 \end{aligned}$$

Therefore, the absolute minimum value of h on $-6 \leq x \leq 5$ is $h(5) = -2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' shifts from negative to positive at $x = 3$

2-8-8

- the slope of h' shifts from negative to positive at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative minimum exists on the graph of h' at $x = 3$

Does not earn the point:

- the concavity at $x = 3$ changes. [Must appeal to the change in sign of h'']
- h' changes from negative to positive
- the graph changes from negative to positive
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = 3$
- the acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is decreasing for $x < 0$ and $x > 0$
- h' is decreasing through $x = 0$
- h'' is always negative before and after $x = 0$
- the slope of h' is always negative before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change. [Must appeal to the lack of change in sign of h'']
- h' is always negative
- h' does not change sign

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of h because h' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -3} (2h'(x) - 1) = 0$ and $\lim_{x \rightarrow -3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -3$

Earns the point:

· $\lim_{x \rightarrow -3} (2h'(x) - 1) = 0$ and $\lim_{x \rightarrow -3} (x^2 - 9) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} \rightarrow \frac{0}{0}$

· $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow -3} \frac{2h'(x)}{2x}$

· $\frac{2h'(-3)}{2(-3)}$

· $\frac{2h'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $\frac{1}{18}$ or equivalent



0	1	2	3
---	---	---	---

2-8-8

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -3$. $\Rightarrow \lim_{x \rightarrow -3} h'(x) = h'(-3)$ and $\lim_{x \rightarrow -3} h''(x) = h''(-3)$.

$$\lim_{x \rightarrow -3} 2h'(x) - 1 = 2 \cdot \frac{1}{2} - 1 = 0$$

$$\lim_{x \rightarrow -3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \lim_{x \rightarrow -3} \frac{2h''(x)}{2x}$$

$$= \frac{h''(-3)}{-3} = \frac{-\frac{1}{6}}{-3} = \frac{1}{18}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $g'(x) = 2x \cdot h'(x) + 2 \cdot h(x)$

Product rule may appear as $g'(3) = 6 \cdot h'(3) + 2 \cdot h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $g'(3) = -20$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = 2x \cdot h'(x) + 2h(x)$$

$$g'(3) = 2 \cdot 3 \cdot h'(3) + 2h(3) = 6 \cdot (-4) + 2 \cdot 2 = -20$$

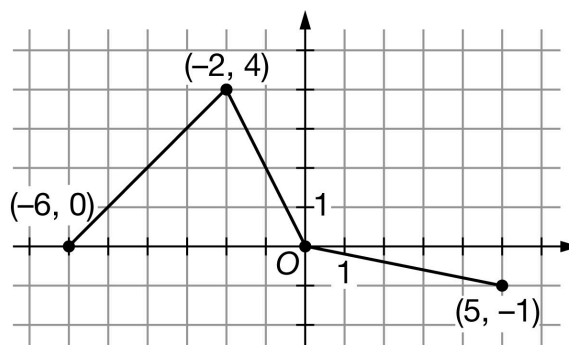
2-8-8

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-6) = -1$.

- Find the x -coordinate of each critical point of g on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute minimum value of g on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -1} \frac{g'(x)-2}{x^2-1}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = 3x \cdot g(x)$. It is known that $g(-2) = 7$. Find the value of $h'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-6 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.

2-8-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $g'(x)$ changes from positive to negative at $x = 0$
- the slope of g changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- g is increasing to the left of $x = 0$ and decreasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for g because $g'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

2-8-8

Earns the first point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-6) = -1$ in calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the absolute minimum value of -1 .

The ordered pair $(-6, -1)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-6	-1
0	11
5	8.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of g on $-6 \leq x \leq 5$ must occur at an endpoint.

$$g(-6) = -1$$

$$g(5) = g(-6) + \int_{-6}^5 g'(t) dt$$

$$= -1 + \int_{-6}^0 g'(t) dt + \int_0^5 g'(t) dt$$

$$= -1 + \frac{1}{2} \cdot 6 \cdot 4 - \frac{1}{2} \cdot 5 \cdot 1 = \frac{17}{2}$$

Therefore, the absolute minimum value of g on $-6 \leq x \leq 5$ is $g(-6) = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = -2$
- the slope of g' changes from positive to negative at $x = -2$
- g'' changes sign at $x = -2$

2-8-8

- a relative extrema exists on the graph of g' at $x = -2$
- a relative maximum exists on the graph of g' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ OR $g''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of g' at $x = -2$
- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is decreasing for $x < 0$ and $x > 0$
- g' is decreasing through $x = 0$
- g'' is always negative before and after $x = 0$
- the slope of g' is always negative before and after $x = 0$
- g'' does not change sign at $x = 0$.
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always negative
- g' does not change sign



0	1
---	---

2-8-8

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of g because g' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -1} (g'(x) - 2) = 0$ and $\lim_{x \rightarrow -1} (x^2 - 1) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -1$

Earns the point:

· $\lim_{x \rightarrow -1} (g'(x) - 2) = 0$ and $\lim_{x \rightarrow -1} (x^2 - 1) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} \rightarrow 0$

· $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow -1} \frac{g'(x)}{2x}$

· $\frac{g''(-1)}{2(-1)}$

· $\frac{g'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer 1 or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Limits of numerator and denominator equal 0

☐ Application of L'Hospital's Rule

2-8-8

☐ Answer**Solution:**

g' and g'' are continuous at $x = -1$. $\Rightarrow \lim_{x \rightarrow -1} g'(x) = g'(-1)$ and $\lim_{x \rightarrow -1} g''(x) = g''(-1)$.

$$\lim_{x \rightarrow -1} g'(x) - 2 = 2 - 2 = 0$$

$$\lim_{x \rightarrow -1} (x^2 - 1) = 1^2 - 1 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} &= \lim_{x \rightarrow -1} \frac{g''(x)}{2x} \\ &= \frac{g''(-1)}{2 \cdot (-1)} = \frac{-2}{-2} = 1 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $h'(x) = 3x \cdot g'(x) + 3 \cdot g(x)$

Product rule may appear as $h'(-2) = -6 \cdot g'(-2) + 3 \cdot g(-2)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(-2) = -3$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

Solution:

$$h'(x) = 3x \cdot g'(x) + 3g(x)$$

$$h'(-2) = 3 \cdot (-2) \cdot g'(-2) + 3g(-2) = -6 \cdot 4 + 3 \cdot 7 = -3$$

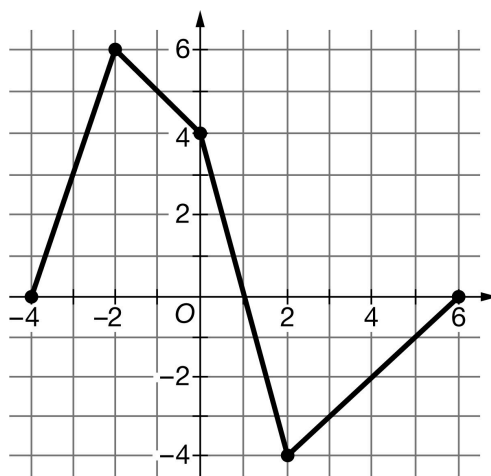
2-8-8

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be a continuous function defined on the closed interval $-4 \leq x \leq 6$. The graph of f , consisting of four line segments, is shown above. Let G be the function defined by $G(x) = \int_0^x f(t) \, dt$.

- On what open intervals is the graph of G concave up? Give a reason for your answer.
- Let P be the function defined by $P(x) = G(x) \cdot f(x)$. Find $P'(3)$.
- Find $\lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x}$.
- Find the average rate of change of G on the interval $[-4, 2]$. Does the Mean Value Theorem guarantee a value c , $-4 < c < 2$, for which $G'(c)$ is equal to this average rate of change? Justify your answer.

Global

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This “global point” can be earned in any one part. Expressions that show this connection and therefore earn this point include: $G' = f$, $G'(x) = f(x)$, $G''(x) = f'(x)$ in part (a), $G'(3) = f(3)$ in part (b), or $G'(2) = f(2)$ in part (c).



0

1

The student response accurately includes the criteria below.

☐ $G'(x) = f(x)$

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

2-8-8

Intervals may also include one or both endpoints.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$$G'(x) = f(x)$$

The graph of G is concave up for $-4 < x < -2$ and $2 < x < 6$, because $G' = f$ is increasing on these intervals.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for the correct application of the product rule in terms of x or in the evaluation of $P'(3)$. Once earned, this point cannot be lost.

The second point is earned by correctly evaluating $G(3) = -3.5$, $G'(3) = -3$, or $f(3) = -3$.

To be eligible to earn the third point a response must have earned the first two points.

Simplification of the numerical value is not required to earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ $G(3)$ or $G'(3)$
- ☐ Answer

Solution:

$$P'(x) = G(x) \cdot f'(x) + f(x) \cdot G'(x)$$

$$P'(3) = G(3) \cdot f'(3) + f(3) \cdot G'(3)$$

Substituting $G(3) = \int_0^3 f(t) dt = -3.5$ and $G'(3) = f(3) = -3$ into the above expression for $P'(3)$ gives the following:

$$P'(3) = -3.5 \cdot 1 + (-3) \cdot (-3) = 5.5$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must show $\lim_{x \rightarrow 2} (x^2 - 2x) = 0$ and $\lim_{x \rightarrow 2} G(x) = 0$ and must show a ratio of the two derivatives, $G'(x)$ and $2x - 2$. The ratio may be shown as evaluations of the derivatives at $x = 2$, such as $\frac{G'(2)}{2}$.

2-8-8

To earn the second point the response must evaluate correctly with appropriate limit notation. In particular the response must include

$$\lim_{x \rightarrow 2} \frac{G'(x)}{2x-2} \text{ or } \lim_{x \rightarrow 2} \frac{f(x)}{2x-2}.$$

With any linkage errors (such as $\frac{G'(x)}{2x-2} = \frac{f(2)}{2}$), the response does not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Uses L'Hospital's Rule
- ☐ Answer with justification

Solution:

$$\lim_{x \rightarrow 2} (x^2 - 2x) = 0$$

Because G is continuous for $-4 \leq x \leq 6$, $\lim_{x \rightarrow 2} G(x) = \int_0^2 f(t) dt = 0$.

Therefore, the limit $\lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x}$ is an indeterminate form of type $\frac{0}{0}$.

Using L'Hospital's Rule,

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{G(x)}{x^2 - 2x} &= \lim_{x \rightarrow 2} \frac{G'(x)}{2x - 2} \\ &= \lim_{x \rightarrow 2} \frac{f(x)}{2x - 2} = \frac{f(2)}{2} = \frac{-4}{2} = -2 \end{aligned}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present at least a difference and a quotient and a correct evaluation. For example, $\frac{0+16}{6}$ or $\frac{G(2)-G(-4)}{6} = \frac{16}{6}$.

Simplification of the numerical value is not required to earn the first point, but any simplification must be correct.

The second point can be earned without the first point if the response has the correct setup but an incorrect or no evaluation of the average rate of change. The Mean Value Theorem need not be explicitly stated provided the conditions and conclusion are stated.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Average rate of change
- ☐ Answer with justification

Solution:

2-8-8

$$G(2) = \int_0^2 f(t) \, dt = 0 \text{ and } G(-4) = \int_0^{-4} f(t) \, dt = -16$$

$$\text{Average rate of change} = \frac{G(2) - G(-4)}{2 - (-4)} = \frac{0 - (-16)}{6} = \frac{8}{3}$$

Yes, $G'(x) = f(x)$ so G is differentiable on $(-4, 2)$ and continuous on $[-4, 2]$. Therefore, the Mean Value Theorem applies and guarantees a value c , $-4 < c < 2$, such that $G'(c) = \frac{8}{3}$.

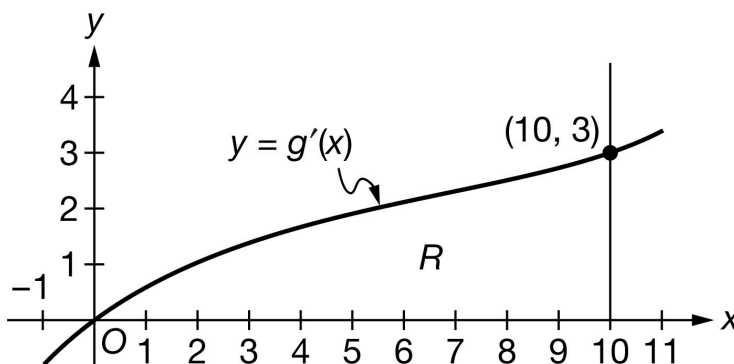
2-8-8

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 11$. The graph of g' has no horizontal tangent lines on the interval $-1 < x < 11$. Let R be the region in the first quadrant bounded by the graph of g' , the x -axis, and the vertical line $x = 10$. It is known that $g(0) = -3$, $g(10) = 15$, and $\int_0^{10} g(x) \, dx = 38$.

- How many points of inflection does the graph of g have on the interval $-1 < x < 11$? Give a reason for your answer.
- Find the average value of g' over the interval $0 \leq x \leq 10$. Show the computations that lead to your answer.
- Must there exist a value of c , for $0 < c < 10$, such that $g(c) = 0$? Justify your answer.
- Evaluate $\int_0^{10} (4g(x) + 5) \, dx$. Show the computations that lead to your answer.
- Evaluate $\lim_{x \rightarrow 10} \frac{g(x) - x + 2}{\sin x}$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = e^x \cdot \int_{10}^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.
- The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by $\int_0^{10} A(x) \, dx$. Write an expression for $A(x)$.
- What is the volume of the solid described in part (g)? Show the computations that lead to your answer.
- Find the value of $\int_0^{10} \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “none,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the increasing behavior of g' .

An appeal to g'' must address the consideration of both $g'' = 0$ and g'' undefined.

2-8-8

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' is increasing on the interval $-1 < x < 11$.

Therefore, the graph of g has no points of inflection on the interval $-1 < x < 11$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(10) - g(0)$.

The answer point is earned only for the correct answer. The values of 15 and -3 must be shown.

$\frac{1}{10}(15 - (-3))$ earns both points.

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\begin{aligned}\text{Average value} &= \frac{1}{10-0} \int_0^{10} g'(x) \, dx = \frac{1}{10}(g(10) - g(0)) \\ &= \frac{1}{10}(15 - (-3)) = 1.8\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

2-8-8

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-3 < 0 < 15$ or $g(0) < 0 < g(10)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.
 $\Rightarrow g$ is continuous on $0 \leq x \leq 10$.

$$g(0) = -3 < 0 < 15 = g(10)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 10$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$4 \cdot 38 + 5 \cdot 10$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Properties of definite integrals
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{10} (4g(x) + 5) dx &= 4 \int_0^{10} g(x) dx + \int_0^{10} 5 dx \\ &= 4 \cdot 38 + 5 \cdot 10 = 202 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{15-10+2}{\sin 10}$. This alone would earn the point.

2-8-8

The student does not have to evaluate $\sin 10$.

An answer of just “ $\frac{7}{\sin 10}$ ” does not earn the point because it does not show the minimal computations.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 10} g(x) = g(10)$

$$\lim_{x \rightarrow 10} \frac{g(x) - x + 2}{\sin x} = \frac{g(10) - 10 + 2}{\sin 10} = \frac{15 - 10 + 2}{\sin 10} = \frac{7}{\sin 10}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-38) + 1 \cdot (-3)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = e^x \cdot \int_{10}^x g(t) dt + e^x \cdot g(x)$$

$$h'(0) = e^0 \cdot \int_{10}^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-38) + 1 \cdot (-3) = -41$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

2-8-8

So, $\int_0^{10} A(x) dx = \int_0^{10} x \cdot g'(x) dx$ earns the point.

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $10 \cdot 15 - 0 - 38$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Integration by parts

☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^{10} x \cdot g'(x) dx = x \cdot g(x) \Big|_0^{10} - \int_0^{10} g(x) dx$$

$$= 10 \cdot g(10) - 0 \cdot g(0) - \int_0^{10} g(x) dx$$

$$= 10 \cdot 15 - 0 \cdot (-3) - 38 = 112$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

2-8-8

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral
- ☐ Answer

Solution:

Let $u = g'(x)$. Then $du = g''(x) dx$.

When $x = 0$, $u = g'(0) = 0$.

When $x = 10$, $u = g'(10) = 3$.

$$\int_0^{10} \frac{g''(x)}{\sqrt{g'(x)}} dx = \int_0^3 \frac{1}{\sqrt{u}} du$$

$$= \lim_{b \rightarrow 0^+} \int_b^3 \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_b^3 = 2\sqrt{3} - \lim_{b \rightarrow 0^+} 2\sqrt{b}$$

$$= 2\sqrt{3} - 0 = 2\sqrt{3}$$

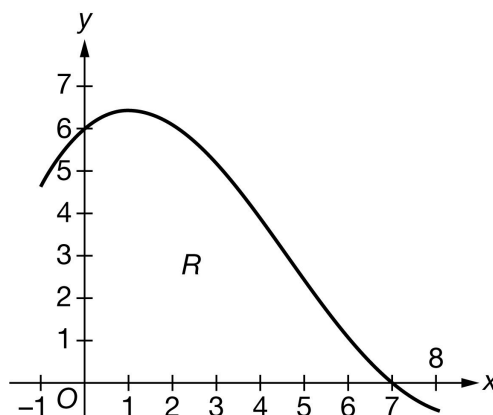
2-8-8

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 8$. The graph of g' has exactly one horizontal tangent line, at $x = 1$. Let R be the region in the first quadrant bounded by the graph of g' and the x -axis from $x = 0$ to $x = 7$. It is known that $g(0) = -15$, $g(7) = 13$, and $\int_0^7 g(x) \, dx = 22.4$.

(a) How many points of inflection does the graph of g have on the interval $-1 < x < 8$? Give a reason for your answer.

(b) Find the average value of g' over the interval $0 \leq x \leq 7$. Show the computations that lead to your answer.

(c) Must there exist a value of c , for $0 < c < 7$, such that $g(c) = 0$? Justify your answer.

(d) Evaluate $\int_0^7 (2g(x) + 3) \, dx$. Show the computations that lead to your answer.

(e) Evaluate $\lim_{x \rightarrow 0} \frac{g(x) + 4x - 4}{\sin x + 1}$. Show the computations that lead to your answer.

(f) Let h be the function defined by $h(x) = e^x \cdot \int_7^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.

(g) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by $\int_0^7 A(x) \, dx$. Write an expression for $A(x)$.

(h) What is the volume of the solid described in part (g)? Show the computations that lead to your answer.

(i) Find the value of $\int_0^7 \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “one,” with some relevant mathematical discussion.

2-8-8

The reasoning point is most commonly earned by correctly appealing to the change in behavior of g' at $x = 1$.

An appeal to g'' must address the consideration of both $g'' = 0$ and g'' undefined.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' changes from increasing to decreasing at $x = 1$.

Therefore, the graph of g has one point of inflection on the interval $-1 < x < 8$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(7) - g(0)$.

The answer point is earned only for the correct answer. The values of 13 and -15 must be shown.

$\frac{1}{7}(13 - (-15))$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\begin{aligned}\text{Average value} &= \frac{1}{7-0} \int_0^7 g'(x) \, dx = \frac{1}{7}(g(7) - g(0)) \\ &= \frac{1}{7}(13 - (-15)) = 4\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

2-8-8

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-15 < 0 < 13$ or $g(0) < 0 < g(7)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous on $0 \leq x \leq 7$.

$$g(0) = -15 < 0 < 13 = g(7)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 7$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$2 \cdot 22.4 + 3 \cdot 7$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Properties of definite integrals
- ☐ Answer

Solution:

$$\int_0^7 (2g(x) + 3) dx = 2 \int_0^7 g(x) dx + \int_0^7 3 dx$$

$$= 2 \cdot 22.4 + 3 \cdot 7 = 65.8$$

Part E

2-8-8

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{-15+0-4}{\sin 0+1}$ or $\frac{-15+0-4}{0+1}$. Either of these alone would earn the point.

The student does not have to evaluate 0.

An answer of just “−19” does not earn the point because it does not show the minimal computations.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 0} g(x) = g(0)$

$$\lim_{x \rightarrow 0} \frac{g(x)+4x-4}{\sin x+1} = \frac{g(0)+0-4}{\sin 0+1} = \frac{-15+0-4}{0+1} = -19$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-22.4) + 1 \cdot (-15)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = e^x \cdot \int_7^x g(t) dt + e^x \cdot g(x)$$

$$h'(0) = e^0 \cdot \int_7^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-22.4) + 1 \cdot (-15) = -37.4$$

2-8-8

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

So, $\int_0^7 A(x) dx = \int_0^7 x \cdot g'(x) dx$ earns the point.

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $7 \cdot 13 - 0 - 22.4$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Integration by parts

☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^7 x \cdot g'(x) dx = x \cdot g(x) \Big|_0^7 - \int_0^7 g(x) dx$$

$$= 7 \cdot g(7) - 0 \cdot g(0) - \int_0^7 g(x) dx$$

$$= 7 \cdot 13 - 0 \cdot (-15) - 22.4 = 68.6$$

2-8-8

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral
- ☐ Answer

Solution:

Let $u = g'(x)$. Then $du = g''(x) dx$.

When $x = 0$, $u = g'(0) = 6$.

When $x = 7$, $u = g'(7) = 0$.

$$\int_0^7 \frac{g''(x)}{\sqrt{g'(x)}} dx = \int_6^0 \frac{1}{\sqrt{u}} du$$

$$= \lim_{b \rightarrow 0^+} \int_6^b \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_6^b = \lim_{b \rightarrow 0^+} 2\sqrt{b} - 2\sqrt{6}$$

$$= 0 - 2\sqrt{6} = -2\sqrt{6}$$

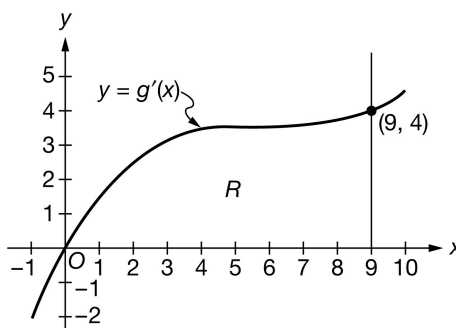
2-8-8

71. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 10$. The graph of g' has exactly one horizontal tangent line, at $x = 6$. Let R be the region in the first quadrant bounded by the graph of g' , the x -axis, and the vertical line $x = 9$. It is known that $g(0) = -10$, $g(9) = 17$, and $\int_0^9 g(x) \, dx = 12.6$.

- How many points of inflection does the graph of g have on the interval $-1 < x < 10$? Give a reason for your answer.
- Find the average value of g' over the interval $0 \leq x \leq 9$. Show the computations that lead to your answer.
- Must there exist a value of c , for $0 < c < 9$, such that $g(c) = 0$? Justify your answer.
- Evaluate $\int_0^9 (5g(x) + 6) \, dx$. Show the computations that lead to your answer.
- Evaluate $\lim_{x \rightarrow 9} \frac{g(x) - 2x + 2}{\cos x}$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = e^x \cdot \int_9^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.
- The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by $\int_0^9 A(x) \, dx$. Write an expression for $A(x)$.
- What is the volume of the solid described in part (g)? Show the computations that lead to your answer.
- Find the value of $\int_0^9 \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “none,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the increasing behavior of g' .

An appeal to g'' must address the consideration of both $g'' = 0$ and g'' undefined.

2-8-8

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' is increasing on the interval $-1 < x < 10$.

Therefore, the graph of g has no points of inflection on the interval $-1 < x < 10$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(9) - g(0)$.

The answer point is earned only for the correct answer. The values of 17 and -10 must be shown.

$\frac{1}{9}(17 - (-10))$ earns both points.

✓

0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\begin{aligned}\text{Average value} &= \frac{1}{9-0} \int_0^9 g'(x) \, dx = \frac{1}{9}(g(9) - g(0)) \\ &= \frac{1}{9}(17 - (-10)) = 3\end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

2-8-8

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-10 < 0 < 17$ or $g(0) < 0 < g(9)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.
 $\Rightarrow g$ is continuous on $0 \leq x \leq 9$.

$$g(0) = -10 < 0 < 17 = g(9)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 9$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$5 \cdot 12.6 + 6 \cdot 9$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Properties of definite integrals
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^9 (5g(x) + 6) dx &= 5 \int_0^9 g(x) dx + \int_0^9 6 dx \\ &= 5 \cdot 12.6 + 6 \cdot 9 = 117 \end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{17-18+2}{\cos 9}$. This alone would earn the point.

2-8-8

The student does not have to evaluate $\cos 9$.

An answer of just “sec 9” does not earn the point because it does not show the minimal computations.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 9} g(x) = g(9)$

$$\lim_{x \rightarrow 9} \frac{g(x) - 2x + 2}{\cos x} = \frac{g(9) - 2 \cdot 9 + 2}{\cos 9} = \frac{17 - 18 + 2}{\cos 9} = \frac{1}{\cos 9}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-12.6) + 1 \cdot (-10)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = e^x \cdot \int_9^x g(t) dt + e^x \cdot g(x)$$

$$h'(0) = e^0 \cdot \int_9^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-12.6) + 1 \cdot (-10) = -22.6$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

2-8-8

So, $\int_0^9 A(x) dx = \int_0^9 x \cdot g'(x) dx$ earns the point.

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $9 \cdot 17 - 0 - 12.6$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Integration by parts

☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^9 x \cdot g'(x) dx = x \cdot g(x) \Big|_0^9 - \int_0^9 g(x) dx$$

$$= 9 \cdot g(9) - 0 \cdot g(0) - \int_0^9 g(x) dx$$

$$= 9 \cdot 17 - 0 \cdot (-10) - 12.6 = 140.4$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

2-8-8

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral
- ☐ Answer

Solution:

Let $u = g'(x)$. Then $du = g''(x) dx$.

When $x = 0$, $u = g'(0) = 0$.

When $x = 9$, $u = g'(9) = 4$.

$$\int_0^9 \frac{g''(x)}{\sqrt{g'(x)}} dx = \int_0^4 \frac{1}{\sqrt{u}} du$$

$$= \lim_{b \rightarrow 0^+} \int_b^4 \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_b^4 = 2\sqrt{4} - \lim_{b \rightarrow 0^+} 2\sqrt{b}$$

$$= 4 - 0 = 4$$

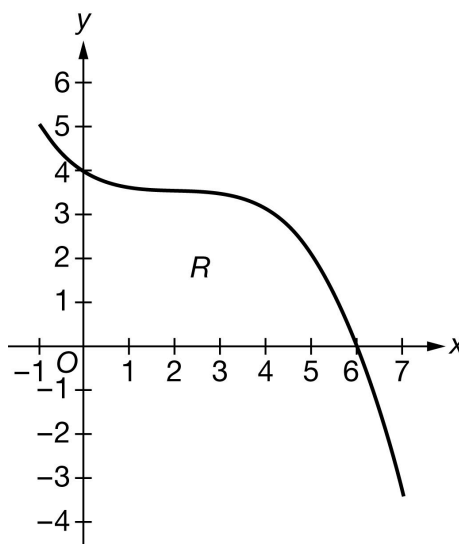
2-8-8

72. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 7$. The graph of g' has exactly one horizontal tangent line, at $x = 2$. Let R be the region in the first quadrant bounded by the graph of g' and the x -axis from $x = 0$ to $x = 6$. It is known that $g(0) = -5$, $g(6) = 13$, and $\int_0^6 g(x) \, dx = 32.4$.

- How many points of inflection does the graph of g have on the interval $-1 < x < 7$? Give a reason for your answer.
- Find the average value of g' over the interval $0 \leq x \leq 6$. Show the computations that lead to your answer.
- Must there exist a value of c , for $0 < c < 6$, such that $g(c) = 0$? Justify your answer.

(d) Evaluate $\int_0^6 (5g(x) + 2) \, dx$. Show the computations that lead to your answer.

(e) Evaluate $\lim_{x \rightarrow 0} \frac{g(x) - 3x + 1}{\cos x + 1}$. Show the computations that lead to your answer.

(f) Let h be the function defined by $h(x) = e^x \cdot \int_6^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.

(g) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by $\int_0^6 A(x) \, dx$. Write an expression for $A(x)$.

(h) What is the volume of the solid described in part (g)? Show the computations that lead to your answer.

(i) Find the value of $\int_0^6 \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-8-8

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “none,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the decreasing behavior of g' .

An appeal to g'' prime must address the consideration of both $g'' = 0$ and g'' undefined.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' is decreasing on the interval $-1 < x < 7$.

Therefore, the graph of g has no points of inflection on the interval $-1 < x < 7$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(6) - g(0)$.

The answer point is earned only for the correct answer. The values of 13 and -5 must be shown.

$\frac{1}{6}(13 - (-5))$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\text{Average value} = \frac{1}{6-0} \int_0^6 g'(x) \, dx = \frac{1}{6}(g(6) - g(0))$$

$$= \frac{1}{6}(13 - (-5)) = 3$$

Part C

2-8-8

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

The response that mentions continuity without establishing continuity is still eligible for the second point.

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-5 < 0 < 13$ or $g(0) < 0 < g(6)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous on $0 \leq x \leq 6$.

$$g(0) = -5 < 0 < 13 = g(6)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 6$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$5 \cdot 32.4 + 2 \cdot 6$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Properties of definite integrals
- ☐ Answer

Solution:

$$\int_0^6 (5g(x) + 2) dx = 5 \int_0^6 g(x) dx + \int_0^6 2 dx$$

2-8-8

$$= 5 \cdot 32.4 + 2 \cdot 6 = 174$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The minimal supporting computations required are $\frac{-5-0+1}{\cos 0+1}$ or $\frac{-5-0+1}{1+1}$. Either of these alone would earn the point.

The student does not have to evaluate $\cos 0$.

An answer of just “−2” does not earn the point because it does not show the minimal computations.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 0} g(x) = g(0)$

$$\lim_{x \rightarrow 0} \frac{g(x) - 3x + 1}{\cos x + 1} = \frac{g(0) - 0 + 1}{\cos 0 + 1} = \frac{-5 - 0 + 1}{1 + 1} = -2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-32.4) + 1 \cdot (-5)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = e^x \cdot \int_6^x g(t) dt + e^x \cdot g(x)$$

2-8-8

$$h'(0) = e^0 \cdot \int_6^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-32.4) + 1 \cdot (-5) = -37.4$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

So, $\int_0^6 A(x) dx = \int_0^6 x \cdot g'(x) dx$ earns the point.

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $6 \cdot 13 - 0 - 32.4$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Integration by parts

☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^6 x \cdot g'(x) dx = x \cdot g(x) \Big|_0^6 - \int_0^6 g(x) dx$$

2-8-8

$$= 6 \cdot g(6) - 0 \cdot g(0) - \int_0^6 g(x) \, dx$$

$$= 6 \cdot 13 - 0 \cdot (-5) - 32.4 = 45.6$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.

✓

0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral
- ☐ Answer

Solution:

Let $u = g'(x)$. Then $du = g''(x) \, dx$.

When $x = 0$, $u = g'(0) = 4$.

When $x = 6$, $u = g'(6) = 0$.

$$\int_0^6 \frac{g''(x)}{\sqrt{g'(x)}} \, dx = \int_4^0 \frac{1}{\sqrt{u}} \, du$$

$$= \lim_{b \rightarrow 0^+} \int_4^b \frac{1}{\sqrt{u}} \, du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_4^b = \lim_{b \rightarrow 0^+} 2\sqrt{b} - 2\sqrt{4}$$

$$= 0 - 4 = -4$$

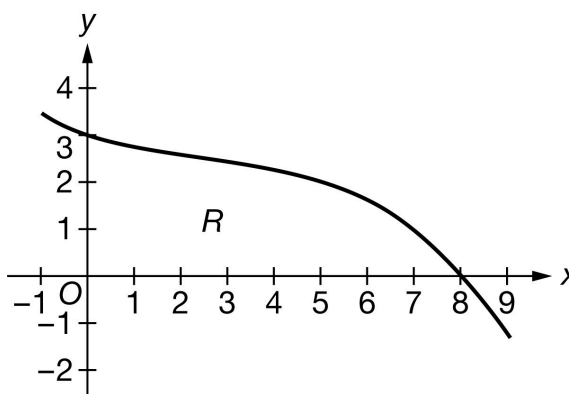
2-8-8

73. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of g' , the derivative of the twice-differentiable function g , is shown above for $-1 < x < 9$. The graph of g' has no horizontal tangent lines on the interval $-1 < x < 9$. Let R be the region in the first quadrant bounded by the graph of g' and the x -axis from $x = 0$ to $x = 8$. It is known that $g(0) = -6$, $g(8) = 10$, and $\int_0^8 g(x) \, dx = 28.8$.

- How many points of inflection does the graph of g have on the interval $-1 < x < 9$? Give a reason for your answer.
- Find the average value of g' over the interval $0 \leq x \leq 8$. Show the computations that lead to your answer.
- Must there exist a value of c , for $0 < c < 8$, such that $g(c) = 0$? Justify your answer.
- Evaluate $\int_0^8 (2g(x) + 5) \, dx$. Show the computations that lead to your answer.
- Evaluate $\lim_{x \rightarrow 0} \frac{g(x) + x - 1}{\sin x - 1}$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = e^x \cdot \int_8^x g(t) \, dt$. Find $h'(0)$. Show the computations that lead to your answer.
- The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis is a rectangle with height x and base in the region R . The volume of the solid is given by $\int_0^8 A(x) \, dx$. Write an expression for $A(x)$.
- What is the volume of the solid described in part (g)? Show the computations that lead to your answer.
- Find the value of $\int_0^8 \frac{g''(x)}{\sqrt{g'(x)}} \, dx$ or show that it does not exist. (Note: $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

We are looking for the application of the relationship between the increasing and decreasing behavior of the first derivative and the points of inflection of the original function.

The first point is earned for “none,” with some relevant mathematical discussion.

The reasoning point is most commonly earned by correctly appealing to the decreasing behavior of g' .

2-8-8

An appeal to g'' must address the consideration of both $g'' = 0$ and g'' undefined.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

The graph of g has a point of inflection where g' changes from increasing to decreasing or from decreasing to increasing.

g' is decreasing on the interval $-1 < x < 9$.

Therefore, the graph of g has no points of inflection on the interval $-1 < x < 9$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Make sure the response addresses the average value of g' , not of g .

The Fundamental Theorem of Calculus point is earned for the expression $g(8) - g(0)$.

The answer point is earned only for the correct answer. The values of 10 and -6 must be shown.

$\frac{1}{8}(10 - (-6))$ earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\text{Average value} = \frac{1}{8-0} \int_0^8 g'(x) \, dx = \frac{1}{8}(g(8) - g(0))$$

$$= \frac{1}{8}(10 - (-6)) = 2$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The student must establish that g is a continuous function. The most common, and acceptable, response is “differentiable implies continuous.”

The student must convey that differentiability implies continuity. Stating g is differentiable and continuous, does not earn the first point.

2-8-8

The response that mentions continuity without establishing continuity is still eligible for the second point.

Using, or citing, the Mean Value Theorem will not earn the second point.

To earn the second point the student must explicitly convey an appropriate inequality involving 0, either mathematically or in words. For example, $-6 < 0 < 10$ or $g(0) < 0 < g(8)$.

The second point requires an answer of “yes” or an equivalent statement.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Intermediate Value Theorem

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous on $0 \leq x \leq 8$.

$$g(0) = -6 < 0 < 10 = g(8)$$

By the Intermediate Value Theorem, there exists a value of c , for $0 < c < 8$, such that $g(c) = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Some (incorrect) responses claim that evaluation of the integrand is equal to the value of the integral.

The first point requires demonstration of both the sum of integrals and the multiplication by a constant.

$2 \cdot 28.8 + 5 \cdot 8$ earns both points. This is the minimal supporting work required.



0	1	2
---	---	---

The student response accurately includes **one** of the criteria below.

- ☐ Properties of definite integrals
- ☐ Answer

Solution:

$$\int_0^8 (2g(x) + 5) dx = 2 \int_0^8 g(x) dx + \int_0^8 5 dx$$

$$= 2 \cdot 28.8 + 5 \cdot 8 = 97.6$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-8-8

The minimal supporting computations required are $\frac{-6-1}{\sin 0-1}$ or $\frac{-6-1}{0-1}$. Either of these alone would earn the point.

The student does not have to evaluate $\sin 0$.

An answer of just “7” does not earn the point because it does not show the minimal computations.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

g is twice differentiable. $\Rightarrow g$ is differentiable.

$\Rightarrow g$ is continuous. $\Rightarrow \lim_{x \rightarrow 0} g(x) = g(0)$

$$\lim_{x \rightarrow 0} \frac{g(x)+x-1}{\sin x-1} = \frac{g(0)+0-1}{\sin 0-1} = \frac{-6+0-1}{0-1} = 7$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

A response of only $e^x \cdot g(x)$ earns only the Fundamental Theorem of Calculus point.

Some responses may reverse the limits of integration and state this is equal to the original integral. This incorrect approach could earn only the first two points, but will not earn the answer point, unless the student resolves the limits reversal issue.

The answer point can only be earned if both of the first two points are earned, and the given answer is correct.

$1 \cdot (-28.8) + 1 \cdot (-6)$ earns all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$h'(x) = e^x \cdot \int_8^x g(t) dt + e^x \cdot g(x)$$

$$h'(0) = e^0 \cdot \int_8^0 g(t) dt + e^0 \cdot g(0) = 1 \cdot (-28.8) + 1 \cdot (-6) = -34.8$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-8-8

If the student writes the integral with $x \cdot g'(x)$ as the integrand, we need to see that it is equal to the integral with $A(x)$ for the integrand.

So, $\int_0^8 A(x) dx = \int_0^8 x \cdot g'(x) dx$ earns the point.

Some responses introduce another variable, or function, for example, $R(x) = g'(x)$, before writing the final expression in terms of this other variable. This earns the point. For example, $A(x) = x \cdot R(x)$ in the presence of $R(x) = g'(x)$.

The point is not earned if the correct integrand appears only in part (h).



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$A(x) = x \cdot g'(x)$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If there is no integration by parts, no points are earned in this part.

The integration by parts point is for an “indefinite integration by parts.” The bounds are associated with the answer point.

Minimal work required to earn the integration by parts point: $8 \cdot 10 - 0 = 28.8$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Integration by parts

☐ Answer

Solution:

$$u = x \quad dv = g'(x) dx$$

$$du = dx \quad v = g(x)$$

$$\int_0^8 x \cdot g'(x) dx = x \cdot g(x) \Big|_0^8 - \int_0^8 g(x) dx$$

$$= 8 \cdot g(8) - 0 \cdot g(0) - \int_0^8 g(x) dx$$

$$= 8 \cdot 10 - 0 \cdot (-6) - 28.8 = 51.2$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-8-8

Many students will write the antiderivative without any evidence of substitution. This antiderivative must be completely correct to earn the substitution point.

Limit notation must be used in order to earn the improper integral point.

The student must earn the improper integral point to be eligible for the answer point.

If the student uses an explicit substitution and makes an error in the limits, they cannot earn the answer point.

A response can earn the substitution point for working only with an indefinite integral.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Improper integral
- ☐ Answer

Solution:

Let $u = g'(x)$. Then $du = g''(x) dx$.

When $x = 0$, $u = g'(0) = 3$.

When $x = 8$, $u = g'(8) = 0$.

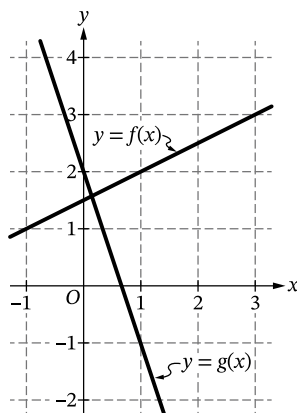
$$\int_0^8 \frac{g''(x)}{\sqrt{g'(x)}} dx = \int_3^0 \frac{1}{\sqrt{u}} du$$

$$= \lim_{b \rightarrow 0^+} \int_3^b \frac{1}{\sqrt{u}} du = \lim_{b \rightarrow 0^+} 2\sqrt{u} \Big|_3^b = \lim_{b \rightarrow 0^+} 2\sqrt{b} - 2\sqrt{3}$$

$$= 0 - 2\sqrt{3} = -2\sqrt{3}$$

2-8-8

74. The graphs of the linear functions f and g are shown in the figure.



If $h(x) = f(x) \cdot g(x)$, then $h'(1) =$

- (A) $-\frac{13}{2}$
(B) $-\frac{3}{2}$
(C) 4
(D) $\frac{11}{2}$

**Answer A**

Correct. If $h(x) = f(x) \cdot g(x)$, then $h'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$ by the product rule. The values of the derivatives at $x = 1$ are determined by the slopes of the lines for the graphs of f and g .

$$f'(1) = \frac{f(3) - f(1)}{3 - 1} = \frac{3 - 2}{2} = \frac{1}{2}$$

$$g'(1) = \frac{g(1) - g(0)}{1 - 0} = \frac{-1 - 2}{1} = -3$$

$$\text{Therefore } h'(1) = f'(1) \cdot g(1) + f(1) \cdot g'(1) = \frac{1}{2} \cdot (-1) + 2 \cdot (-3) = -\frac{1}{2} - 6 = -\frac{13}{2}.$$

2-8-8

75. A particle starts at time $t = 0$ and moves along the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = (t - 1)^3(2t - 3)$.

- Find the velocity of the particle at any time $t \geq 0$.
- For what values of t is the velocity of the particle less than zero?
- Find the value of t when the particle is moving and the acceleration is zero.

Part A

2 points are earned for the derivative

1 point is **deducted** for algebra error

$$\begin{aligned} v(t) &= x'(t) \\ &= 3(t - 1)^2(2t - 3) + 2(t - 1)^3 \\ &= (t - 1)^2(8t - 11) \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for the derivative

1 point is **deducted** for algebra error

$$\begin{aligned} v(t) &= x'(t) \\ &= 3(t - 1)^2(2t - 3) + 2(t - 1)^3 \\ &= (t - 1)^2(8t - 11) \end{aligned}$$

Part B

1 point is earned for setting their $v(t) < 0$

3 points are earned for answer

1 point is **deducted** for allowing $t = 1$

1 point is **deducted** for allowing $t < 0$

1 point is **deducted** for including $t = \frac{11}{8}$

1 point is **deducted** for algebra error

$$v(t) < 0 \text{ when } (t - 1)^2(8t - 11) < 0$$

$$\therefore 8t - 11 < 0 \text{ and } t \neq 1$$

$$\text{or } t < \frac{11}{8} \text{ and } t \neq 1$$

2-8-8

Since $t \geq 0$, answer is

$$0 \leq t < \frac{11}{8}, \text{ except } t = 1$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for setting their $v(t) < 0$

3 points are earned for answer

1 point is **deducted** for allowing $t = 1$

1 point is **deducted** for allowing $t < 0$

1 point is **deducted** for including $t = \frac{11}{8}$

1 point is **deducted** for algebra error

$$v(t) < 0 \text{ when } (t - 1)^2(8t - 11) < 0$$

$$\therefore 8t - 11 < 0 \text{ and } t \neq 1$$

$$\text{or } t < \frac{11}{8} \text{ and } t \neq 1$$

Since $t \geq 0$, answer is

$$0 \leq t < \frac{11}{8}, \text{ except } t = 1$$

Part C

1 point is earned for derivative of their v from (a) or (b)

1 point is earned for solving for $v'(t) = 0$ and solves for at least 2 roots

1 point is earned for answer

$$a(t) = v'(t)$$

$$= 2(t - 1)(8t - 11) + 8(t - 1)^2$$

$$= 6(t - 1)(4t - 5)$$

$$a(t) = 0 \text{ when } t = 1, t = \frac{5}{4}$$

2-8-8

but particle not moving at $t = 1$ so $t = \frac{5}{4}$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for derivative of their v from (a) or (b)

1 point is earned for solving for $v'(t) = 0$ and solves for at least 2 roots

1 point is earned for answer

$$a(t) = v'(t)$$

$$= 2(t - 1)(8t - 11) + 8(t - 1)^2$$

$$= 6(t - 1)(4t - 5)$$

$$a(t) = 0 \text{ when } t = 1, t = \frac{5}{4}$$

but particle not moving at $t = 1$ so $t = \frac{5}{4}$

2-8-8

76. A particle moves on the x -axis so that its position at any time $t \geq 0$ is given by $x(t) = 2te^{-t}$.

- (a) Find the acceleration of the particle at $t = 0$.
- (b) Find the velocity of the particle when its acceleration is 0.
- (c) Find the total distance traveled by the particle from $t = 0$ to $t = 5$.

Part A

1 point is earned for correctly finding $x'(t)$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

1 point is earned for compute $x''(0)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$

$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$

$$a(0) = x''(0) = -2 - 2 = -4$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly finding $x'(t)$

1 point is earned for correctly finding $x''(t)$

1 point is earned for correctly identifying $x''(t)$ with acceleration

1 point is earned for compute $x''(0)$

$$v(t) = x'(t) = 2e^{-t} - 2te^{-t}$$

$$a(t) = x''(t) = -2e^{-t} - 2e^{-t} + 2te^{-t}$$

$$a(0) = x''(0) = -2 - 2 = -4$$

Part B

1 point is earned for correctly solving $x''(t) = 0$

1 point is earned for correctly computing $x'(t_0)$ for all solutions found

$$x''(t) = -2e^{-t}(2 - t) = 0$$

$$t = 2$$

$$x'(2) = v(2) = 2e^{-2}(1 - 2)$$

$$= \frac{-2}{e^2}$$

$$\approx -0.271$$



0	1	2
---	---	---

2-8-8

The student response earns all of the following points:

1 point is earned for correctly solving $x''(t) = 0$

1 point is earned for correctly computing $x'(t_0)$ for all solutions found

$$\begin{aligned}x''(t) &= -2e^{-t}(2 - t) = 0 \\t &= 2 \\x'(2) &= v(2) = 2e^{-2}(1 - 2) \\&= \frac{-2}{e^2} \\&\approx -0.271\end{aligned}$$

Part C

1 point is earned for correctly solving $x'(t) = 0$ or otherwise identifying turning point

1 point is earned for correctly evaluating $x(0)$, $x(5)$, and $x(\text{turning point})$

1 point is earned for the correct answer

Note: Max 1/3 if no turning point or turning point ignored

$$\begin{aligned}x'(t) &= v(t) = 2e^{-t}(1 - t) = 0 \\t &= 1 \\x(0) &= 0 \\x(1) &= \frac{2}{e} \approx 0.736 \\x(5) &= \frac{10}{e^5} \approx 0.067\end{aligned}$$

$$\begin{aligned}\text{Distance} &= \frac{2}{e} + \frac{2}{e} - \frac{10}{e^5} \\&\approx 1.404\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correctly solving $x'(t) = 0$ or otherwise identifying turning point

1 point is earned for correctly evaluating $x(0)$, $x(5)$, and $x(\text{turning point})$

1 point is earned for the correct answer

Note: Max 1/3 if no turning point or turning point ignored

2-8-8

$$x'(t) = v(t) = 2e^{-t}(1 - t) = 0$$

$$t = 1$$

$$x(0) = 0$$

$$x(1) = \frac{2}{e} \approx 0.736$$

$$x(5) = \frac{10}{e^5} \approx 0.067$$

$$\begin{aligned}\text{Distance} &= \frac{2}{e} + \frac{2}{e} - \frac{10}{e^5} \\ &\approx 1.404\end{aligned}$$

77. Let f be a differentiable function such that $f(3) = 4$ and $f'(3) = 5$. If $g(x) = x^2 f(x)$, what is the value of $g'(3)$?

(A) 11

(B) 24

(C) 30

(D) 69

**Answer D**

Correct. The product rule is applied to determine the derivative, as follows.

$$g'(x) = \frac{d}{dx}(x^2)f(x) + x^2 f'(x) = 2xf(x) + x^2 f'(x)$$

$$g'(3) = 2 \cdot 3f(3) + 3^2 f'(3) = 6 \cdot 4 + 9 \cdot 5 = 69$$

78. Let f be a differentiable function such that $f(2) = 2$ and $f'(2) = 5$. If $g(x) = x^3 f(x)$, what is the value of $g'(2)$?

(A) 17

(B) 24

(C) 60

(D) 64

**Answer D**

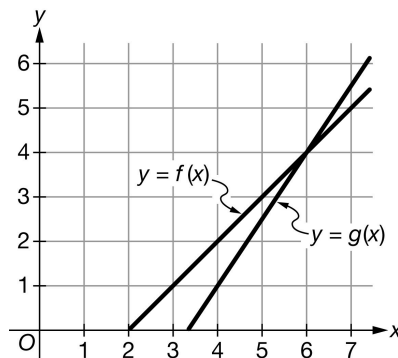
Correct. The product rule is applied to determine the derivative, as follows.

$$g'(x) = \frac{d}{dx}(x^3)f(x) + x^3 f'(x) = 3x^2 f(x) + x^3 f'(x)$$

$$g'(2) = 3 \cdot (2)^2 \cdot f(2) + 2^3 \cdot f'(2) = 12 \cdot 2 + 8 \cdot 5 = 24 + 40 = 64$$

2-8-8

79.



The graphs of the linear function f and the linear function g are shown in the figure above. If $h(x) = f(x)g(x)$, then $h'(4) =$

- (A) -2
 (B) $\frac{3}{2}$
 (C) $\frac{7}{2}$
 (D) 4

**Answer D**

Correct. The product rule is applied to determine that $h'(x) = f'(x)g(x) + f(x)g'(x)$. The values of $f'(4)$ and $g'(4)$ are determined from the slopes of the graphs of f and g at $x = 4$ and substituted into the expression for $h'(4)$, as follows.

$$h'(4) = f'(4)g(4) + f(4)g'(4) = 1 \cdot 1 + 2 \cdot \left(\frac{3}{2}\right) = 1 + 3 = 4$$

80. If $f(x) = \ln x \cos x$, then $f'(x) =$

- (A) $-\frac{\sin x}{x}$
 (B) $\frac{\sin x}{x}$
 (C) $\frac{\cos x}{x} - \ln x \sin x$
 (D) $\frac{\cos x}{x} + \ln x \sin x$

**Answer C**

Correct. The product rule is applied to determine the derivative, as follows.

$$f'(x) = \frac{d}{dx}(\ln x) \cdot \cos x + \ln x \cdot \frac{d}{dx}(\cos x) = \frac{1}{x} \cdot \cos x + \ln x \cdot (-\sin x) = \frac{\cos x}{x} - \ln x \cdot \sin x$$

2-8-8

81.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
4	2	7	3	5

The table above gives the values of the differentiable functions f and g and their derivatives at $x = 4$. What is the value of $\frac{d}{dx}(f(x)g(x))$ at $x = 4$?

- (A) 11
(B) 29
(C) 31
(D) 35

**Answer C**

Correct. The product rule is applied and correct values are chosen from the table, as follows.

$$\left. \frac{d}{dx}(f(x)g(x)) \right|_{x=4} = f'(4)g(4) + f(4)g'(4) = 7 \cdot 3 + 2 \cdot 5 = 31$$

82. The function f is given by $f(x) = (x^3 + bx + 6)g(x)$, where b is a constant and g is a differentiable function satisfying $g(2) = 3$ and $g'(2) = -1$. For what value of b is $f'(2) = 0$?

- (A) -7
(B) -10
(C) -12
(D) -22

**Answer D**

Correct. Use the product rule to calculate $f'(x)$, then evaluate at $x = 2$ and set equal to 0, as follows.

$$f'(x) = (3x^2 + b)g(x) + (x^3 + bx + 6)g'(x)$$

$$f'(2) = (12 + b)(3) + (8 + 2b + 6)(-1) = 22 + b = 0 \Rightarrow b = -22$$

2-8-8

83.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	2	-4	-5	3
2	-3	1	8	4

The table above gives values of the differentiable functions f and g and their derivatives at selected values of x . If h is the function defined by $h(x) = f(x)g(x) + 2g(x)$, then $h'(1) =$

(A) 32

(B) 30

(C) -6

(D) -16

Answer A

$$f'(1)g(1) + f(1)g'(1) + 2g'(1)$$

84. If $f'(x) = x^2$ and $g'(x) = 3e^x$ and if $h(x) = f(x)g(x)$, then $h'(x)$ could be

(A) $3x^2e^x$ (B) $x^3e^x + 3x^2e^x$ (C) $x^3e^x + x^2e^x$ (D) $3x^2e^x + 6xe^x$ **Answer B**

Correct. By the product rule, $h'(x) = f(x)g'(x) + f'(x)g(x)$. Because $f'(x) = x^2$ and $g'(x) = 3e^x$, one possibility is $f(x) = \frac{1}{3}x^3$ and $g(x) = 3e^x$. With these choices, $h'(x) = \left(\frac{1}{3}x^3\right)(3e^x) + (x^2)(3e^x) = x^3e^x + 3x^2e^x$.

85. If $f(x) = \ln(x) \cos(4x)$, then $f'(x) =$

(A) $-\frac{4 \sin(4x)}{x}$ (B) $\frac{\cos(4x)}{x} - \ln(x) \sin(4x)$ (C) $\frac{\cos(4x)}{x} - 4 \ln(x) \sin(4x)$ (D) $\frac{\cos(4x)}{x} + 4 \ln(x) \sin(4x)$ **Answer C**

Correct. Both the product rule and the chain rule must be used to compute the derivative.

$$\begin{aligned} f'(x) &= \left(\frac{d}{dx} \ln(x)\right) \cdot \cos(4x) + \ln(x) \cdot \left(\frac{d}{dx} \cos(4x)\right) = \frac{1}{x} \cdot \cos(4x) + \ln(x) \cdot (-4 \sin(4x)) \\ &= \frac{\cos(4x)}{x} - 4 \ln(x) \sin(4x) \end{aligned}$$

2-8-8

86. If $g(x) = x^2\sqrt{x+1}$, then $g'(3) =$

(A) $\frac{3}{2}$

(B) $\frac{39}{4}$

(C) $\frac{57}{4}$

(D) 36

**Answer C**Correct. The derivative of g is found using the product rule.

$$g'(x) = 2x \cdot \sqrt{x+1} + x^2 \cdot \frac{1}{2\sqrt{x+1}}$$

$$g'(3) = 6(2) + \frac{9}{2(2)} = 12 + \frac{9}{4} = \frac{57}{4}$$

87. If $f(x) = e^x \sin x$, then $f'(x) =$

(A) $e^x \cos x$

(B) $-e^x \cos x$

(C) $e^x(\sin x + \cos x)$

(D) $e^x(\sin x - \cos x)$

**Answer C**

Correct. The product rule is applied to determine the derivative, as follows.

$$f'(x) = \frac{d}{dx}(e^x) \cdot \sin x + e^x \cdot \frac{d}{dx}(\sin x) = e^x \cdot \sin x + e^x \cdot \cos x = e^x(\sin x + \cos x)$$

88.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	1	5	2	4

The table above gives the values of the differentiable functions f and g and their derivatives at $x = 2$. What is the value of $\frac{d}{dx}(f(x)g(x))$ at $x = 2$?

(A) 6

(B) 13

(C) 14

(D) 20

**Answer C**

Correct. The product rule is applied and correct values are chosen from the table, as follows.

$$\frac{d}{dx}(f(x)g(x))\bigg|_{x=2} = f'(2)g(2) + f(2)g'(2) = 5 \cdot 2 + 1 \cdot 4 = 14$$

2-8-8

89. The function f is given by $f(x) = (2x^3 + bx)g(x)$, where b is a constant and g is a differentiable function satisfying $g(2) = 4$ and $g'(2) = -1$. For what value of b is $f'(2) = 0$?

- (A) -8
(B) $-\frac{56}{3}$
(C) -24
(D) -40

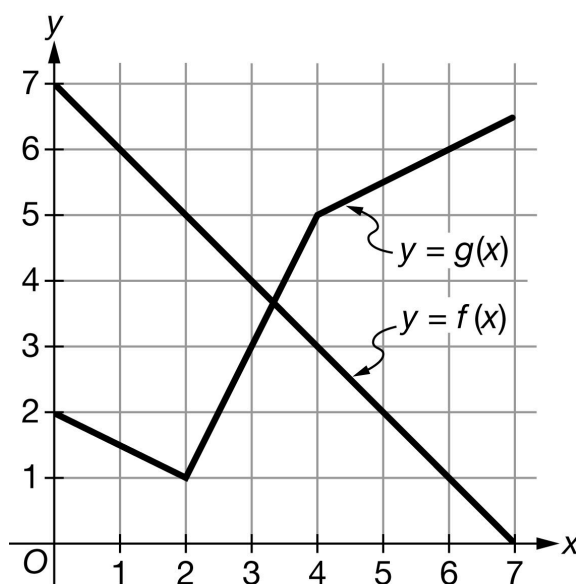
**Answer D**

Correct. Use the product rule to calculate $f'(x)$, then evaluate at $x = 2$ and set equal to 0, as follows.

$$f'(x) = (6x^2 + b)g(x) + (2x^3 + bx)g'(x)$$

$$f'(2) = (24 + b)(4) + (16 + 2b)(-1) = 2b + 80 = 0 \Rightarrow b = -40$$

90.



The graphs of the linear function f and the piecewise linear function g are shown in the figure above. If $h(x) = f(x)g(x)$, then $h'(3) =$

- (A) -11
(B) -2
(C) 2
(D) 5

**Answer D**

Correct. The product rule is applied to determine that $h'(x) = f'(x)g(x) + f(x)g'(x)$. The values of $f'(3)$ and $g'(3)$ are determined from the slopes of the graphs of f and g at $x = 3$ and substituted into the expression for $h'(3)$, as follows.

$$h'(3) = f'(3)g(3) + f(3)g'(3) = (-1) \cdot 3 + 4 \cdot 2 = 5$$

2-8-8

91. If $f(x) = \sqrt{x} \cos x$, then $f'(x) =$

(A) $-\frac{\sin x}{2\sqrt{x}}$

(B) $\frac{1-2\sqrt{x} \sin x}{2\sqrt{x}}$

(C) $\frac{\cos x - 2x \sin x}{2\sqrt{x}}$ ✓

(D) $\frac{\cos x + 2x \sin x}{2\sqrt{x}}$

Answer C

Correct. The product rule is applied, as follows:

$$f'(x) = \frac{d}{dx}(\sqrt{x}) \cos x + \sqrt{x} \frac{d}{dx}(\cos x) = \frac{1}{2\sqrt{x}} \cos x + \sqrt{x}(-\sin x) = \frac{\cos x - 2x \sin x}{2\sqrt{x}}.$$

92. If $f(x) = \frac{1}{x} \cdot \cos x$, then $f'(x) =$

(A) $\frac{\sin x}{x^2}$

(B) $\frac{-1-x^2 \sin x}{x^2}$

(C) $\frac{-\cos x - x \sin x}{x^2}$ ✓

(D) $\frac{-\cos x + x \sin x}{x^2}$

Answer C

Correct. The product rule is applied as follows.

$$f'(x) = \frac{d}{dx}\left(\frac{1}{x}\right) \cos x + \frac{1}{x} \frac{d}{dx}(\cos x) = \frac{-\cos x}{x^2} - \frac{\sin x}{x} = \frac{-\cos x - x \sin x}{x^2}$$

93. The base of a solid is the region in the first quadrant bounded by the y -axis, the x -axis, the graph of $y = e^x$, and the vertical line $x = 1$. For this solid, each cross section perpendicular to the x -axis is a square.

What is the volume of the solid?

(A) $e - 1$

(B) $\frac{1}{2}e^2 - \frac{1}{2}$ ✓

(C) $e^2 - 1$

(D) $2e^2 - 2$

Answer B

Correct. The area of a square of side length s is s^2 . A typical cross section of the solid is a square with side from the x -axis to the graph of $y = e^x$. The length of the side of the square is therefore $s = e^x$, so the area of the square is $(e^x)^2 = e^{2x}$. The volume of the solid is found using the definite integral of the cross-sectional area.

$$\int_0^1 e^{2x} dx = \frac{1}{2} e^{2x} \Big|_0^1 = \frac{1}{2} e^2 - \frac{1}{2}$$

2-8-8

94. Let f be a continuous function defined on the interval $0 \leq x \leq 7$. The function f and its derivatives have the properties indicated in the table shown, where DNE indicates the x -values at which the derivatives of f do not exist.

x	$x = 0$	$0 < x < 2$	$x = 2$	$2 < x < 4$	$x = 4$	$4 < x < 7$	$x = 7$
$f(x)$	-1	Negative	-4	Negative	-2	Negative	0
$f'(x)$	DNE	Negative	DNE	Positive	0	Positive	DNE
$f''(x)$	DNE	Positive	DNE	Negative	0	Positive	DNE

Part A

On what open intervals contained in $0 < x < 7$, if any, is the graph of f both increasing and concave down? Give a reason for your answer.

Part B

Find the average rate of change of f over the interval $0 \leq x \leq 4$. Can the Mean Value Theorem be applied to f on the interval $0 \leq x \leq 4$ to guarantee a value of c , for $0 < c < 4$, such that $f'(c)$ is equal to this average rate of change? Give a reason for your answer.

Part C

Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^7 f(x) dx$.

Part D

Let g be the function defined by $g(x) = x \cdot f(x)$. Is g increasing or decreasing at $x = 1$? Give a reason for your answer.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned only by an answer of $2 < x < 4$ or this interval including one or both endpoints.
- Special case: A response of “ f is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there” earns **P1**.
- Special case: A response of “ f is concave down on $2 < x < 4$ because $f''(x) < 0$ there” earns **P1**.

Model Solution:

f is increasing and concave down on the interval $2 < x < 4$.



0	1
---	---

The response accurately includes the criteria below.

- Answer

2-8-8

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P1** to be eligible for **P2**.
- To earn **P2**, a response must correctly cite both $f'(x) > 0$ and $f''(x) < 0$.
- Special case: A response of “ f is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there” does not earn **P2**.
- Special case: A response of “ f is concave down on $2 < x < 4$ because $f''(x) < 0$ there” does not earn **P2**.

Model Solution:

The graph of function f is increasing on an interval when $f'(x) > 0$ and concave down on an interval when $f''(x) < 0$.



0	1
---	---

The response accurately includes the criteria below.

- Reason

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P3**, a response must include a difference and a quotient. Any of the following is sufficient to earn **P3**: $\frac{f(4)-f(0)}{4-0} = -\frac{1}{4}$, $\frac{-2-(-1)}{4}$, or $\frac{-1}{4-0}$.

Model Solution:

$$\frac{f(4)-f(0)}{4-0} = \frac{-2-(-1)}{4} = -\frac{1}{4}$$



0	1
---	---

The response accurately includes the criteria below.

- Average rate of change

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response does not need to have earned **P3** to be eligible for **P4**.
- To earn **P4**, a response must note that f is not differentiable and conclude “no.”
- To earn **P4**, a response need not explicitly name the Mean Value Theorem, but if a theorem is named, it must be correct.

Model Solution:

2-8-8

No, the Mean Value Theorem cannot be applied to f on the interval $0 \leq x \leq 4$ because f is not differentiable at $x = 2$.



0	1
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The response accurately includes the criteria below.

- Answer with reason

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- Read “=” as “ $\approx$ ” for **P5**.
- The form of a left Riemann sum includes three terms, each of which includes a product of two factors. To earn **P5**, at least five of the six factors must be correct.
- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ earns **P5**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P5**.
- An unsupported answer of -16 does not earn **P5**.
- A completely correct right Riemann sum with accompanying work (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) earns **P5**. A response with any errors or missing factors in the right Riemann sum does not earn **P5**.

Model Solution:

$$\int_0^7 f(x)dx \approx f(0)(2-0) + f(2)(4-2) + f(4)(7-4)$$



0	1
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The response accurately includes the criteria below.

- Form of left Riemann sum

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ is eligible to earn **P6** with correct values for $f(0)$, $f(2)$, and $f(4)$ imported from the table. If any of the six factors is incorrect, the response does not earn **P6**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P6**.
- A response that includes $(-1)(2) + (-4)(2) + (-2)(3)$ or equivalent banks **P6** (i.e., subsequent errors in simplification will not be considered in scoring for **P6**).
- An unsupported answer of -16 does not earn **P6**.
- A right Riemann sum (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) does not earn **P6**.

Model Solution:

$$(-1)(2) + (-4)(2) + (-2)(3) = -16$$

2-8-8



0	1
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The response accurately includes the criteria below.

- Answer with supporting work

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for considering either $g'(x)$ or $g'(1)$.

Model Solution:

$$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$$

$$g'(1) = f(1) + 1 \cdot f'(1)$$



0	1
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The response accurately includes the criteria below.

- Considers $g'(x)$

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P8** is earned with either $g'(x) = f(x) + x \cdot f'(x)$ or $g'(1) = f(1) + 1 \cdot f'(1)$.

Model Solution:

$$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$$

$$g'(1) = f(1) + 1 \cdot f'(1)$$



0	1
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The response accurately includes the criteria below.

- $g'(x)$ or $g'(1)$.

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

2-8-8

- A response must have earned **P7** and **P8** to be eligible for **P9**.
- An eligible response that states either “decreasing because $g'(1) < 0$ ” or “decreasing because $g'(x) < 0$ for $0 < x < 2$ ” earns **P9**.
- A response that states “ $g(1)$ is decreasing” does not earn **P9**.
- A response that estimates negative values for $f(1)$ and/or $f'(1)$ using the table and reaches a correct conclusion earns **P9**.

Model Solution:

Because $g'(1) < 0$ and g' is continuous on $0 < x < 2$, g is decreasing on an interval containing $x = 1$.



0	1
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The response accurately includes the criteria below.

- Answer with reason